

Supplementary material for: “Sparse Bayesian factor analysis when the number of factors is unknown”

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Appendix A: Initializing Algorithm 1

Algorithm 1 is based on initial values for r , δ_r , $\mathbf{f}_r$ and r_{sp} . To check the mixing of the MCMC chain, two (or more) independent runs with different initial dimensions r of the CFA model and a positive number r_{sp} of spurious columns are performed. On the one hand, r is chosen to be smaller than the expected number of factors, on the other hand a large value close to $H - r_{sp}$ is chosen.

Initial values for the pivots $\mathbf{l}_r = (l_1, \dots, l_r)$ of the sparsity matrix δ_r are chosen by first sampling l_1 from $\{1, \dots, u_1\}$, where u_1 is a small number, e.g. 5. Then for $j = 2, \dots, r$, we sample l_j uniformly from the set $\mathcal{L}(\mathbf{l}_{-j})$ with $\mathbf{l}_{-j} = \{l_1, \dots, l_{j-1}\}$. Factor loadings below the pivots are allowed to be zero with positive probability p_0 , e.g. $p_0 = 0.5$. We draw at most 100 initial values δ_r (including the pivots) in this way, until a matrix δ_r is obtained which satisfies the 3579 counting rule. If no such sparsity matrix is found, then we add enough non-zero elements in each column of δ_r (e.g., by setting $\delta_{l_j+1,j} = 1, \dots, \delta_{l_j+3,j} = 1$) to ensure variance identification for the initial value.

To obtain starting values for the factors $\mathbf{f}_r$, we perform a few (say 100) MCMC iterations in the confirmatory factor model corresponding to δ_r , which is initialized by sampling the factors $\mathbf{f}_r = (\mathbf{f}_1^r, \dots, \mathbf{f}_T^r)$ from the prior: $f_{jt} \sim \mathcal{N}(0, 1)$, $j = 1, \dots, r$, $t = 1, \dots, T$. While holding δ_r fixed, we iterate between sampling the model parameters β_r and $\sigma_1^2, \dots, \sigma_m^2$ as in Step (P) and sampling the factors $\mathbf{f}_r$ as in Step (F) of Algorithm 1. The resulting factors $\mathbf{f}_r$ serve as starting values for the full-blown MCMC scheme described in Algorithm 1.

Appendix B: Details on Step (H)

In this section, we provide details on updating the hyperparameters α and γ for the 2PB prior (3.3) without conditioning on the slab probabilities $\tau_1, \dots, \tau_H$. Inference is performed conditionally on the current column sizes $d_1, \dots, d_H$ and the number of

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spurious columns r_{sp} in the EFA model and the current pivots $\mathbf{l}_r = (l_1, \dots, l_r)$ in the CFA model:

$$p(\alpha, \gamma | \cdot) \propto p(\alpha)p(\gamma) \prod_{j:d_j > 1} \Pr(\boldsymbol{\delta}_{:,j}^H | \alpha, \gamma, l_j) \prod_{j:d_j \leq 1} \Pr(d_j | \alpha, \gamma),$$

where $\alpha \sim \mathcal{G}(a^\alpha, b^\alpha)$, $\gamma \sim \mathcal{G}(a^\gamma, b^\gamma)$ and $\boldsymbol{\delta}_{:,j}^H$ is the j th columns of $\boldsymbol{\delta}_H$.

The conditional posterior $p(\alpha, \gamma | \cdot)$ is based on the likelihood (3.8) for active columns, the likelihood (F.2) for spurious columns⁴ and the likelihood (F.1) for zero columns introduced in Appendix F:

$$p(\alpha, \gamma | \cdot) \propto \frac{p(\alpha)p(\gamma)}{B(\frac{\alpha\gamma}{H}, \gamma)^H} \cdot B(\frac{\alpha\gamma}{H}, \gamma + m - r - r_{sp})^{H-r-r_{sp}} \\ \cdot \prod_{j:d_j > 1} B(\frac{\alpha\gamma}{H} + d_j - 1, \gamma + m - l_j - d_j + 1) \cdot \prod_{j_{sp}=1}^{r_{sp}} B(\frac{\alpha\gamma}{H} + 1, \gamma + m - r - j_{sp}).$$

MH-steps based on a random walk proposal for, respectively, $\log \alpha$ and $\log \gamma$ are implemented to sample from $p(\alpha | \gamma, \cdot)$ and $p(\gamma | \alpha, \cdot)$.

Appendix C: Details on Step (P)

Step (P) of Algorithm 1 updates the parameters in the confirmatory sparse factor model

$$\mathbf{f}_t^r \sim \mathcal{N}_r(\mathbf{0}, \mathbf{I}_r), \quad \mathbf{y}_t = \boldsymbol{\beta}_r \mathbf{f}_t^r + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}_m(\mathbf{0}, \boldsymbol{\Sigma}_r), \quad \boldsymbol{\Sigma}_r = \text{Diag}(\sigma_1^2, \dots, \sigma_m^2), \quad (\text{C.1})$$

where the indicator matrix $\boldsymbol{\delta}_r$ imposes a certain zero structure on the loading matrix $\boldsymbol{\beta}_r$. The joint posterior distribution $p(\boldsymbol{\beta}_i^\delta, \sigma_i^2 | \mathbf{y}, \mathbf{f}_r, \boldsymbol{\delta}_r)$ of the non-zero factor loadings $\boldsymbol{\beta}_i^\delta$ and the idiosyncratic variance σ_i^2 is derived for each row i ($i = 1, \dots, m$) conditional on the factors $\mathbf{f}_r = (\mathbf{f}_1^r, \dots, \mathbf{f}_T^r)$ and the indicator matrix $\boldsymbol{\delta}_r$ from the following regression model:

$$\tilde{\mathbf{y}}_i = \mathbf{X}_i^\delta \boldsymbol{\beta}_i^\delta + \tilde{\boldsymbol{\epsilon}}_i, \quad (\text{C.2})$$

where $\tilde{\mathbf{y}}_i = (y_{i1} \cdots y_{iT})'$ and $\tilde{\boldsymbol{\epsilon}}_i = (\epsilon_{i1} \cdots \epsilon_{iT})' \sim \mathcal{N}_T(0, \sigma_i^2 \mathbf{I})$. $\mathbf{X}_i^\delta$ is a regressor matrix for $\boldsymbol{\beta}_i^\delta$ constructed from the $T \times r$ dimensional latent factor matrix $\mathbf{F} = (\mathbf{f}_1^r \cdots \mathbf{f}_T^r)'$ in the following way. If no element in row i of $\boldsymbol{\beta}_r$ is restricted to 0, then $\mathbf{X}_i^\delta = \mathbf{F}$. If some elements are restricted to 0, then $\mathbf{X}_i^\delta$ is obtained from $\mathbf{F}$ by deleting all columns j where $\delta_{ij} = 0$, i.e. $\mathbf{X}_i^\delta = \mathbf{F} \boldsymbol{\Pi}_i^\delta$, where $\boldsymbol{\Pi}_i^\delta$ is a $r \times \sum_{j=1}^r \delta_{ij}$ selection matrix, selecting those columns j of $\mathbf{F}$ where $\delta_{ij} \neq 0$.

C.1 Prior distributions in a confirmatory factor model

The likelihood derived from (C.2) is combined either with a hierarchical Gaussian prior or a fractional prior $\boldsymbol{\beta}_i^\delta | \sigma_i^2$ for the vector $\boldsymbol{\beta}_i^\delta$ of unconstrained elements in row i . A

⁴Note that in (F.2), r_{sp} is the number of spurious columns excluding column j_{sp} . Hence, the likelihood is computed sequentially from the first to the last spurious column.

hierarchical Gaussian prior takes the form

$$\beta_{i\cdot}^\delta | \sigma_i^2 \sim \mathcal{N}_{q_i}(\mathbf{0}, \mathbf{B}_{i0}^\delta \sigma_i^2), \quad (\text{C.3})$$

where $\mathbf{B}_{i0}^\delta$ is a diagonal matrix derived from the various priors (3.10), (3.11), and (3.12) introduced in Section 3.2.

As an alternative shrinkage prior, Frühwirth-Schnatter and Lopes (2010) introduced a conditionally conjugate fractional prior $p(\beta_{i\cdot}^\delta | \sigma_i^2, b, \mathbf{f}_r) \propto p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \beta_{i\cdot}^\delta, \sigma_i^2)^b$ in the spirit of O'Hagan (1995), see also Frühwirth-Schnatter and Lopes (2018). It can be interpreted as the posterior of a non-informative prior and a small fraction $b > 0$ of the conditional likelihood derived from regression model (C.2). This yields:

$$\beta_{i\cdot}^\delta | \sigma_i^2, b, \mathbf{f}_r \sim \mathcal{N}_{q_i}(\mathbf{b}_{iT}^\delta, \mathbf{B}_{iT}^\delta \sigma_i^2 / b), \quad (\text{C.4})$$

where $\mathbf{b}_{iT}^\delta = \mathbf{B}_{iT}^\delta (\mathbf{X}_i^\delta)' \tilde{\mathbf{y}}_i$ and $\mathbf{B}_{iT}^\delta = ((\mathbf{X}_i^\delta)' \mathbf{X}_i^\delta)^{-1}$. Similar fractional priors have been applied for variable selection in other latent variable models (Frühwirth-Schnatter and Tüchler, 2008; Frühwirth-Schnatter and Wagner, 2010).

Both priors can be combined either with the inverse gamma prior (3.14) or the F-prior (3.15) on $\sigma_1^2, \dots, \sigma_m^2$ to derive the joint posterior $p(\beta_{i\cdot}^\delta, \sigma_i^2 | \mathbf{y}, \mathbf{f}_r, \delta_r)$, see Section C.2. For the inverse gamma prior (3.14), $C_{i0} = b_i^\sigma$ are fixed hyperparameters for $i = 1, \dots, m$, while the scaling parameters $C_{10}, \dots, C_{m0}$ are random variables for the F-prior (3.15) that are sampled during MCMC from the conditional gamma posterior $C_{i0} | b_i^\sigma, \sigma_i^2$ given in (E.7) for $i = 1, \dots, m$.

As discussed in Section 3.4, the hyperparameters $b_1^\sigma, \dots, b_m^\sigma$ depend on the diagonal elements of estimators $\hat{\Omega}^{-1}$ (Frühwirth-Schnatter and Lopes, 2018) or $\hat{\Omega}$ of either Ω^{-1} or Ω . In Frühwirth-Schnatter and Lopes (2018), Bayesian estimators are derived by combining the sample information with an inverse Wishart prior $\Omega^{-1} \sim \mathcal{W}_m(\nu_o + (m+1)/2, \nu_o \mathbf{S}_o)$ and yields following Wishart posterior distribution:

$$\Omega^{-1} \sim \mathcal{W}_m \left(\nu_o + (m+1)/2 + \frac{T}{2}, \nu_o \mathbf{S}_o + \frac{1}{2} \sum_{t=1}^T \mathbf{y}_t \mathbf{y}_t' \right).$$

assuming that $\mathbf{y}_t$ is demeaned. Optimal estimators both for Ω^{-1} and Ω are derived in Yang and Berger (1994) from such a posterior distribution. The optimal estimator for Ω^{-1} under the entropy loss function L (see Appendix K.2 for a definition of L) is the inverse of the posterior expectation $E(\Omega | \mathbf{y})$ of Ω :

$$\widehat{\Omega^{-1}} = \left(\nu_o + \frac{T}{2} \right) \left(\nu_o \mathbf{S}_o + \frac{T}{2} \sum_{t=1}^T \mathbf{y}_t \mathbf{y}_t' \right)^{-1}. \quad (\text{C.5})$$

The optimal estimator for Ω under the entropy loss function L is the inverse of the posterior expectation $E(\Omega^{-1} | \mathbf{y})$ of Ω^{-1} :

$$\hat{\Omega} = \left(\nu_o \mathbf{S}_o + \frac{T}{2} \sum_{t=1}^T \mathbf{y}_t \mathbf{y}_t' \right) \left(\nu_o + \frac{m+1}{2} + \frac{T}{2} \right)^{-1}. \quad (\text{C.6})$$

Frühwirth-Schnatter and Lopes (2018) recommend to choose $\nu_o = 3$ and $\mathbf{S}_o = \mathbf{I}_m$.

C.2 Posterior distributions in a confirmatory factor model

For a sparse factor model, the precise form of the posterior $p(\beta_i^\delta, \sigma_i^2 | \mathbf{y}, \mathbf{f}_r, \boldsymbol{\delta}_r)$ depends on the number of non-zero elements in the i th row of β_r , i.e. $q_i = \sum_{j=1}^r \delta_{ij}$. There are basically three types of rows, when it comes to updating the parameters: zero rows, dedicated rows and rows with multiple loadings. For zero rows (i.e. $q_i = 0$), (C.2) reduces to a “null” model without regressors $\mathbf{X}_i^\delta$, that is $\tilde{\mathbf{y}}_i = \tilde{\boldsymbol{\epsilon}}_i$. Hence, the posterior of σ_i^2 is simply given by

$$\sigma_i^2 | \tilde{\mathbf{y}}_i, \mathbf{f}_r, \boldsymbol{\delta}_r \sim \mathcal{G}^{-1}(c_T^n, C_{iT}^n), \quad c_T^n = c^\sigma + \frac{T}{2}, \quad C_{iT}^n = C_{i0} + \frac{1}{2} \sum_{t=1}^T y_{it}^2. \quad (\text{C.7})$$

For all non-zero rows (i.e. $q_i > 0$), the posterior of $(\beta_i^\delta, \sigma_i^2)$ in row i is given by:

$$\sigma_i^2 | \tilde{\mathbf{y}}_i, \mathbf{f}_r, \boldsymbol{\delta}_r \sim \mathcal{G}^{-1}(c_T, C_{iT}^\delta), \quad \beta_i^\delta | \sigma_i^2, \tilde{\mathbf{y}}_i, \mathbf{f}_r, \boldsymbol{\delta}_r \sim \mathcal{N}_{q_i}(\mathbf{B}_{iT}^\delta \mathbf{m}_{iT}^\delta, \mathbf{B}_{iT}^\delta \sigma_i^2). \quad (\text{C.8})$$

For the hierarchical Gaussian prior (C.3), the moments are given by:

$$\begin{aligned} (\mathbf{B}_{iT}^\delta)^{-1} &= (\mathbf{B}_{i0}^\delta)^{-1} + (\mathbf{X}_i^\delta)' \mathbf{X}_i^\delta, & \mathbf{m}_{iT}^\delta &= (\mathbf{X}_i^\delta)' \tilde{\mathbf{y}}_i, \\ c_T &= c^\sigma + \frac{T}{2}, & C_{iT}^\delta &= C_{i0} + \frac{1}{2} \text{SSR}_i, \quad \text{SSR}_i = \tilde{\mathbf{y}}_i' \tilde{\mathbf{y}}_i - (\mathbf{m}_{iT}^\delta)' \mathbf{B}_{iT}^\delta \mathbf{m}_{iT}^\delta. \end{aligned} \quad (\text{C.9})$$

For the fractional prior (C.4), the moments are given by:

$$\begin{aligned} (\mathbf{B}_{iT}^\delta)^{-1} &= (\mathbf{X}_i^\delta)' \mathbf{X}_i^\delta, & \mathbf{m}_{iT}^\delta &= (\mathbf{X}_i^\delta)' \tilde{\mathbf{y}}_i, \\ c_T &= c^\sigma + \frac{(1-b)T}{2}, & C_{iT}^\delta &= C_{i0} + \frac{(1-b)}{2} \text{SSR}_i, \end{aligned} \quad (\text{C.10})$$

where SSR_i is the same as in (C.9) and, for the fractional prior, reduces to the residual sum of squared errors.⁵ For dedicated rows (i.e. $q_i = 1$) only a single non-zero factor loading β_{i,j_i} is present in a particular column j_i and the posterior given in (C.8) simplifies considerably:

$$\sigma_i^2 | \tilde{\mathbf{y}}_i, \mathbf{f}_r, \boldsymbol{\delta}_r \sim \mathcal{G}^{-1}(c_T, C_{iT}), \quad \beta_{i,j_i} | \sigma_i^2, \tilde{\mathbf{y}}_i, \mathbf{f}_r, \boldsymbol{\delta}_r \sim \mathcal{N}(B_{iT} m_{iT}, B_{iT} \sigma_i^2). \quad (\text{C.11})$$

For a hierarchical Gaussian prior, the posterior moments are given by:

$$\begin{aligned} B_{iT} &= 1/(B_{i0,j_i j_i}^{-1} + \sum_{t=1}^T f_{j_i,t}^2), & m_{iT} &= \sum_{t=1}^T f_{j_i,t} y_{it}, \\ c_T &= c^\sigma + \frac{T}{2}, & C_{iT} &= C_{i0} + \frac{1}{2} \left(\sum_{t=1}^T y_{it}^2 - m_{iT}^2 B_{iT} \right), \end{aligned} \quad (\text{C.12})$$

where $B_{i0,j_i j_i}$ is the prior variance of the dedicated factor loading in row i , whereas for a fractional prior:

$$\begin{aligned} B_{iT} &= 1/\left(\sum_{t=1}^T f_{j_i,t}^2\right), & m_{iT} &= \sum_{t=1}^T f_{j_i,t} y_{it}, \\ c_T &= c^\sigma + \frac{(1-b)T}{2}, & C_{iT} &= C_{i0} + \frac{(1-b)}{2} \left(\sum_{t=1}^T y_{it}^2 - m_{iT}^2 B_{iT} \right). \end{aligned} \quad (\text{C.13})$$

⁵If the residual $\boldsymbol{\epsilon}_i = \tilde{\mathbf{y}}_i - \mathbf{X}_i^\delta \mathbf{B}_{iT}^\delta \mathbf{m}_{iT}^\delta$ is defined in the usual way, then:

$$\begin{aligned} \boldsymbol{\epsilon}_i' \boldsymbol{\epsilon}_i &= \tilde{\mathbf{y}}_i' \tilde{\mathbf{y}}_i - (\mathbf{m}_{iT}^\delta)' \mathbf{B}_{iT}^\delta (\mathbf{X}_i^\delta)' \tilde{\mathbf{y}}_i - \tilde{\mathbf{y}}_i' \mathbf{X}_i^\delta \mathbf{B}_{iT}^\delta \mathbf{m}_{iT}^\delta + (\mathbf{m}_{iT}^\delta)' \mathbf{B}_{iT}^\delta (\mathbf{X}_i^\delta)' \mathbf{X}_i^\delta \mathbf{B}_{iT}^\delta \mathbf{m}_{iT}^\delta \\ &= \tilde{\mathbf{y}}_i' \tilde{\mathbf{y}}_i - (\mathbf{m}_{iT}^\delta)' \mathbf{B}_{iT}^\delta \mathbf{m}_{iT}^\delta - (\mathbf{m}_{iT}^\delta)' \mathbf{B}_{iT}^\delta \mathbf{m}_{iT}^\delta + (\mathbf{m}_{iT}^\delta)' \mathbf{B}_{iT}^\delta \mathbf{m}_{iT}^\delta = \text{SSR}_i. \end{aligned}$$

C.3 Block sampling of idiosyncratic variances and factor loadings

Step (P) in Algorithm 1 could be implemented as in Lopes and West (2004), by sampling β_i^δ and σ_i^2 from the posterior distribution $p(\beta_i^\delta, \sigma_i^2 | \mathbf{y}, \mathbf{f}_r, \delta_r)$ derived in Section C.2 row by row. However, an important improvement is feasible through block sampling of all idiosyncratic variances and all non-zero factor loadings, summarized in Algorithm C.1.⁶ The use of the Cholesky decomposition of the information matrix (instead of the covariance matrix) to sample from a high-dimensional density is fashioned after Rue and Held (2005, Theorem 2.5 and Algorithm 2.5) who consider Gaussian random fields.

Algorithm C.1. Sampling parameters for a sparse Bayesian factor model

- (P-a) For all zero rows, sample σ_i^2 from (C.7), which can be trivially vectorized.
- (P-b) If the remaining rows are all dedicated with a single non-zero loading in column j_i (which can be different for different rows), then sampling from (C.11) is easily vectorized, since all posterior moments are univariate.
- (P-c) Even if some of the non-zero rows are not dedicated, joint sampling of all idiosyncratic variances and factor loadings is feasible for all non-zero rows. Let $i_1, \dots, i_n$ be the indices of all n non-zero rows of β_r , i.e. $q_{i_l} > 0$ for $l = 1, \dots, n$. Let $d = \sum_i q_i$ be the total number of non-zero elements in β_r and let $\beta_r^\delta = (\beta_{i_1}^\delta \dots \beta_{i_n}^\delta)'$ be the d -dimensional vector obtained by stacking row by row all non-zero elements in each row of β_r . To sample the idiosyncratic variances $\sigma_{i_1}^2, \dots, \sigma_{i_n}^2$ and the non-zero factor loadings β_r^δ in β_r jointly, proceed in the following way:

- (P-c1) Construct the information matrix $\mathbf{P}$ and the covector $\mathbf{m}$ of the joint posterior

$$\beta_r^\delta | \sigma_{i_1}^2, \dots, \sigma_{i_n}^2, \mathbf{f}_r, \mathbf{y} \sim \mathcal{N}_d(\mathbf{P}^{-1} \mathbf{m}, \mathbf{P}^{-1} \mathbf{D}).$$

The matrix $\mathbf{D} = \text{Diag}(\sigma_{i_1}^2 \mathbf{1}_{1 \times q_{i_1}} \dots \sigma_{i_n}^2 \mathbf{1}_{1 \times q_{i_n}})$, with $\mathbf{1}_{1 \times l}$ being a $1 \times l$ row vector of ones, is a $d \times d$ diagonal matrix containing the idiosyncratic variances, while the $d \times d$ matrix $\mathbf{P}$ and the $d \times 1$ vector $\mathbf{m}$ are given by:

$$\mathbf{P} = \begin{pmatrix} (\mathbf{B}_{i_1, T}^\delta)^{-1} & \mathbf{O} & \dots & \mathbf{O} \\ \mathbf{O} & (\mathbf{B}_{i_2, T}^\delta)^{-1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \mathbf{O} \\ \mathbf{O} & \dots & \mathbf{O} & (\mathbf{B}_{i_n, T}^\delta)^{-1} \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} \mathbf{m}_{i_1, T}^\delta \\ \vdots \\ \mathbf{m}_{i_n, T}^\delta \end{pmatrix},$$

where $(\mathbf{B}_{i_l, T}^\delta)^{-1}$ and $\mathbf{m}_{i_l, T}^\delta$ are the information matrix and the covector appearing in the posterior (C.8) of the non-zero elements in row i_l . $\mathbf{P}$ is a sparse band matrix with maximal band width equal to $\max q_{i_l}$.

- (P-c2) Compute the Cholesky decomposition $\mathbf{P} = \mathbf{L}\mathbf{L}'$, where $\mathbf{L}$ is lower triangular, using a special algorithm developed for band matrices. Next, solve $\mathbf{L}\mathbf{x} = \mathbf{m}$ for $\mathbf{x}$ using an algorithm specially designed for triangular matrices. Evidently, $\mathbf{x}$ is a $d \times 1$ vector.

⁶This algorithm has been implemented for the first time in the unpublished research report by Frühwirth-Schnatter and Lopes (2010).

(P-c3) Sample $\sigma_{i_1}^2, \dots, \sigma_{i_n}^2$ jointly from (C.8). The squared sum $\mathbf{x}'\mathbf{x}$ can be used to vectorize the computation of $C_{i_l, T}^\delta$ for each $l = 1, \dots, n$, since

$$\mathbf{x}_{i_l}' \mathbf{x}_{i_l} = (\mathbf{m}_{i_l, T}^\delta)' \mathbf{B}_{i_l, T}^\delta \mathbf{m}_{i_l, T}^\delta, \quad (\text{C.14})$$

where $\mathbf{x}_{i_l}$ is the q_{i_l} -dimensional subvector of $\mathbf{x}$ corresponding to $\beta_{i_l, \cdot}^\delta$.

(P-c4) Finally, define the diagonal matrix $\mathbf{D}$ from $\sigma_{i_1}^2, \dots, \sigma_{i_n}^2$ as described above and draw $\mathbf{z} \sim \mathcal{N}_d(\mathbf{0}, \mathbf{D})$. Solving the system

$$\mathbf{L}' \beta_r^\delta = \mathbf{x} + \mathbf{z} \quad (\text{C.15})$$

for β_r^δ leads to a draw from the joint posterior $\beta_r^\delta | \sigma_{i_1}^2, \dots, \sigma_{i_n}^2, \mathbf{y}, \mathbf{f}_r$.

To derive (C.14), let $\mathbf{L}_{i_l}$ be the $q_{i_l} \times q_{i_l}$ submatrix of $\mathbf{L}$ corresponding to $\beta_{i_l, \cdot}^\delta$. Evidently, $\mathbf{L}_{i_l}$ is equal to the Cholesky decomposition of the individual information matrix $(\mathbf{B}_{i_l, T}^\delta)^{-1}$. Furthermore, the q_{i_l} -dimensional subvector $\mathbf{x}_{i_l}$ corresponding to $\beta_{i_l, \cdot}^\delta$ satisfies $\mathbf{L}_{i_l} \mathbf{x}_{i_l} = \mathbf{m}_{i_l, T}^\delta$. Therefore:

$$\mathbf{x}_{i_l}' \mathbf{x}_{i_l} = (\mathbf{m}_{i_l, T}^\delta)' (\mathbf{L}_{i_l}')^{-1} \mathbf{L}_{i_l}^{-1} \mathbf{m}_{i_l, T}^\delta = (\mathbf{m}_{i_l, T}^\delta)' (\mathbf{L}_{i_l} \mathbf{L}_{i_l}')^{-1} \mathbf{m}_{i_l, T}^\delta = (\mathbf{m}_{i_l, T}^\delta)' \mathbf{B}_{i_l, T}^\delta \mathbf{m}_{i_l, T}^\delta.$$

It is easy to prove that the solution β_r^δ of (C.15) is a draw from the posterior $p(\beta_r^\delta | \sigma_{i_1}^2, \dots, \sigma_{i_n}^2, \mathbf{y}, \mathbf{f})$. Note that $\mathbf{L} \mathbf{L}' \beta_r^\delta = \mathbf{L} \mathbf{x} + \mathbf{L} \mathbf{z} = \mathbf{m} + \mathbf{L} \mathbf{z}$. Therefore

$$\beta_r^\delta = (\mathbf{L} \mathbf{L}')^{-1} \mathbf{m} + (\mathbf{L} \mathbf{L}')^{-1} \mathbf{L} \mathbf{z} = \mathbf{P}^{-1} \mathbf{m} + (\mathbf{L}')^{-1} \mathbf{z}.$$

Evidently, $E(\beta_r^\delta) = \mathbf{P}^{-1} \mathbf{m}$. Since for each $l = 1, \dots, n$, $\mathbf{L}_{i_l} \sigma_{i_l}^2 = \sigma_{i_l}^2 \mathbf{L}_{i_l}$, it holds that $\mathbf{L} \mathbf{D} = \mathbf{D} \mathbf{L}$ and therefore $\mathbf{D} \mathbf{L}^{-1} = \mathbf{L}^{-1} \mathbf{D}$. Since $V(\beta_r^\delta) = (\mathbf{L}')^{-1} \mathbf{D} \mathbf{L}^{-1} = (\mathbf{L}')^{-1} \mathbf{L}^{-1} \mathbf{D} = \mathbf{P}^{-1} \mathbf{D}$, it follows that $\beta_r^\delta \sim \mathcal{N}_d(\mathbf{P}^{-1} \mathbf{m}, \mathbf{P}^{-1} \mathbf{D})$.

Appendix D: Details on Step (D)

D.1 Marginal likelihoods when the factors are known

Although we work throughout this paper with a factor model where the factors $\mathbf{f}_t^r$ are latent, several steps of Algorithm 1 perform model selection with respect to δ_r conditional on the most recent draw of the factors $\mathbf{f}_r = (\mathbf{f}_1^r, \dots, \mathbf{f}_T^r)$ in the confirmatory factor model (C.1). Hence, to sample new indicators δ_i in row i , the marginal likelihood $p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \delta_r)$ of regression model (C.2) is needed.

If δ_i is a zero row (i.e $q_i = 0$), then the marginal likelihood simplifies to

$$p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \delta_r) = p(\tilde{\mathbf{y}}_i) = \frac{\Gamma(c_T^n) (C_{i0})^{c^\sigma}}{(2\pi)^{T/2} \Gamma(c^\sigma) (C_{iT}^n)^{c_T^n}}, \quad (\text{D.1})$$

where c_T^n and C_{iT}^n are the posterior moments of σ_i^2 under the “null” model given by (C.7).

If at least one element of δ_i is different from zero, then the marginal likelihood computation differs between the hierarchical Gaussian prior (C.3) and the fractional prior (C.4).

Marginal likelihoods for a hierarchical Gaussian prior. For a hierarchical Gaussian prior, a well-known exercise in Bayesian regression analysis yields:

$$p(\tilde{\mathbf{y}}_i | \boldsymbol{\delta}_r, \mathbf{f}_r) = \frac{1}{(2\pi)^{T/2}} \frac{|\mathbf{B}_{iT}^\delta|^{1/2}}{|\mathbf{B}_{i0}^\delta|^{1/2}} \frac{\Gamma(c_T)(C_{i0})^{c^\sigma}}{\Gamma(c^\sigma)(C_{iT}^\delta)^{c_T}}, \quad (\text{D.2})$$

where $\mathbf{B}_{iT}^\delta$, c_T and C_{iT}^δ are the posterior moments of $p(\boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2 | \boldsymbol{\delta}_r, \tilde{\mathbf{y}}_i, \mathbf{f}_r)$ given by (C.9).

Marginal likelihoods for a fractional prior. For a fractional prior, the derivation of the marginal likelihood, if at least one element of $\boldsymbol{\delta}_{i\cdot}$ is different from zero, is less standard and can be obtained in a similar way as in Frühwirth-Schnatter and Wagner (2010). A fraction b of the full conditional likelihood of regression model (C.2) is used to define the fractional prior $p(\boldsymbol{\beta}_{i\cdot}^\delta | \sigma_i^2, b, \mathbf{f}_r)$ in (C.4):

$$p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2) = p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2)^{1-b} p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2)^b \propto p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2)^{1-b} p(\boldsymbol{\beta}_{i\cdot}^\delta | \sigma_i^2, b, \mathbf{f}_r).$$

The remaining part of the likelihood, that is $p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2)^{1-b}$, is used for model selection and is combined with the prior $p(\sigma_i^2)$ defined in (3.14) and the *normalized* fractional prior $p(\boldsymbol{\beta}_{i\cdot}^\delta | \sigma_i^2, b, \mathbf{f}_r)$, given by:

$$p(\boldsymbol{\beta}_{i\cdot}^\delta | \sigma_i^2, b, \mathbf{f}_r) = \frac{p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2)^b}{c_i(\sigma_i^2, \mathbf{f}_r, b)}.$$

The normalizing constant $c_i(\sigma_i^2, \mathbf{f}_r, b)$ is given by:

$$c_i(\sigma_i^2, \mathbf{f}_r, b) = \int p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2)^b d\boldsymbol{\beta}_{i\cdot}^\delta = (2\pi\sigma_i^2)^{\frac{q_i - Tb}{2}} b^{-\frac{q_i}{2}} |\mathbf{B}_{iT}^\delta|^{1/2} \exp\left(-\frac{b}{2\sigma_i^2} \text{SSR}_i\right),$$

where $\mathbf{B}_{iT}^\delta$ and SSR_i are the posterior moments of $p(\boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2 | \mathbf{f}_r, \boldsymbol{\delta}_{i\cdot}, \tilde{\mathbf{y}}_i)$ given by (C.10). Integrating the fractional posterior

$$p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2)^{1-b} p(\boldsymbol{\beta}_{i\cdot}^\delta | \sigma_i^2, b, \mathbf{f}_r) p(\sigma_i^2)$$

over $\boldsymbol{\beta}_{i\cdot}^\delta$, yields the fractional likelihood $p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \sigma_i^2, b)$:

$$\begin{aligned} p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \sigma_i^2, b) &= \int p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2)^{1-b} p(\boldsymbol{\beta}_{i\cdot}^\delta | \mathbf{f}_r, \sigma_i^2, b) d\boldsymbol{\beta}_{i\cdot}^\delta \\ &= \frac{1}{c_i(\sigma_i^2, \mathbf{f}_r, b)} \int p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \boldsymbol{\beta}_{i\cdot}^\delta, \sigma_i^2) d\boldsymbol{\beta}_{i\cdot}^\delta \\ &= \left(\frac{1}{2\pi\sigma_i^2}\right)^{\frac{(T-q_i)-(Tb-q_i)}{2}} b^{\frac{q_i}{2}} \frac{|\mathbf{B}_{iT}^\delta|^{1/2}}{|\mathbf{B}_{iT}^\delta|^{1/2}} \exp\left(-\frac{(1-b)}{2\sigma_i^2} \text{SSR}_i\right) \\ &= \left(\frac{1}{2\pi\sigma_i^2}\right)^{\frac{T(1-b)}{2}} b^{\frac{q_i}{2}} \exp\left(-\frac{(1-b)}{2\sigma_i^2} \text{SSR}_i\right). \end{aligned}$$

When we combine $p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \sigma_i^2, b)$ with the prior $p(\sigma_i^2)$, then we obtain:

$$p(\tilde{\mathbf{y}}_i | \mathbf{f}_r, \sigma_i^2, b) p(\sigma_i^2) = \frac{C_{i0}^{c^\sigma}}{\Gamma(c^\sigma)(2\pi)^{T(1-b)/2}} b^{\frac{q_i}{2}} \left(\frac{1}{\sigma_i^2}\right)^{c^\sigma + \frac{T(1-b)}{2}} \exp\left(-\frac{C_{i0} + \text{SSR}_i(1-b)/2}{\sigma_i^2}\right),$$

which is the kernel of the inverse gamma distribution in (C.8). Integrating the right hand side with respect to σ_i^2 yields the marginal likelihood under the fractional prior:

$$p(\tilde{\mathbf{y}}_i | \boldsymbol{\delta}_r, \mathbf{f}_r) = \frac{b^{q_i/2} \Gamma(c_T) (C_{i0})^{c^\sigma}}{(2\pi)^{T(1-b)/2} \Gamma(c^\sigma) (C_{iT}^{\boldsymbol{\delta}})^{c_T}}. \quad (\text{D.3})$$

D.2 Multi-move sampling of a set of indicators in a column

Another important building block of MCMC inference for sparse Bayesian factor models is sampling all indicators δ_{ij} in column j for a set of rows $i \in I_j \subseteq \{1, \dots, m\}$, conditional on the factors $\mathbf{f}_r = (\mathbf{f}_1^r, \dots, \mathbf{f}_T^r)$, the remaining indicators $\boldsymbol{\delta}_{i,-j}^r$ and the hyperparameter τ_j , without conditioning on the model parameters $\boldsymbol{\beta}_r$ and $\sigma_1^2, \dots, \sigma_m^2$, see Step (D-a) of Algorithm 1.

According to the prior (3.2), the indicators δ_{ij} in different rows i are independent apriori conditional on the hyperparameter τ_j , with the log prior odds O_{ij}^{pr} of $\delta_{ij} = 1$ versus $\delta_{ij} = 0$ being given by:

$$O_{ij}^{\text{pr}} = \log \frac{\Pr(\delta_{ij} = 1 | \tau_j)}{\Pr(\delta_{ij} = 0 | \tau_j)} = \log \frac{\tau_j}{1 - \tau_j}. \quad (\text{D.4})$$

To sample δ_{ij} conditional on $\boldsymbol{\delta}_{i,-j}^r$ and $\mathbf{f}_r$, without conditioning on $\boldsymbol{\beta}_r$ and $(\sigma_1^2, \dots, \sigma_m^2)$, the log posterior odds O_{ij}^{post} , given by

$$\begin{aligned} O_{ij}^{\text{post}} &= \log \frac{\Pr(\delta_{ij} = 1 | \boldsymbol{\delta}_{i,-j}, \tau_j, \tilde{\mathbf{y}}_i, \mathbf{f}_r)}{\Pr(\delta_{ij} = 0 | \boldsymbol{\delta}_{i,-j}, \tau_j, \tilde{\mathbf{y}}_i, \mathbf{f}_r)} = \log \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 1, \boldsymbol{\delta}_{i,-j}, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 0, \boldsymbol{\delta}_{i,-j}, \mathbf{f}_r)} + O_{ij}^{\text{pr}} \\ &= O_{ij} + O_{ij}^{\text{pr}}, \end{aligned} \quad (\text{D.5})$$

is required which combines the log prior odds O_{ij}^{pr} given in (D.4) with the log likelihood ratio O_{ij} , given by:

$$O_{ij} = \log \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 1, \boldsymbol{\delta}_{i,-j}, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 0, \boldsymbol{\delta}_{i,-j}, \mathbf{f}_r)}, \quad (\text{D.6})$$

where $\boldsymbol{\delta}_{i,-j}$ are the remaining indicators in row i . The likelihood ratio O_{ij} is easily computed from the marginal likelihoods $p(\tilde{\mathbf{y}}_i | \delta_{ij}, \boldsymbol{\delta}_{i,-j}, \mathbf{f}_r)$ where, respectively, $\delta_{ij} = 1$ and $\delta_{ij} = 0$. As discussed in Section D.1, these marginal likelihoods are available in closed form and may be computed for each row $i \in I_j$ separately for $\delta_{ij} = 0$ and $\delta_{ij} = 1$.

However, this procedure is likely to be inefficient, in particular, if the set I_j is large. To achieve greater efficiency, Algorithm D.1 outlined below provides a technique to compute directly the log likelihood ratio O_{ij} (rather than the individual marginal likelihoods) *simultaneously* for all rows $i \in I_j$. This allows joint sampling of all indicators δ_{ij} in column j for all rows $i \in I_j$.

The precise form of the log likelihood ratio O_{ij} of $\delta_{ij} = 1$ versus $\delta_{ij} = 0$ defined in (D.6) depends on the remaining indicators $\boldsymbol{\delta}_{i,-j}$ in row i . The computation of O_{ij} is easily vectorized for all rows $i \in I_j$ where all elements of $\boldsymbol{\delta}_{i,-j}$ are zero. In this case, a

model where observation y_{it} is dedicated to factor j ($\delta_{ij} = 1$) is compared to a model where y_{it} is uncorrelated with all remaining observations ($\delta_{ij} = 0$). In particular, O_{ij} is easily obtained from the marginal likelihood of a dedicated model with $j_i = j$ and the “null” model. As shown in Algorithm D.1, it is possible (but less straightforward) to vectorize the computation of the log likelihood ratio O_{ij} also for the remaining rows $i \in I_j$ where at least one element of $\delta_{i,-j}$ is different from zero.

Algorithm D.1. Multi-move sampling of indicators in a column. Sample all indicators δ_{ij} in column $\delta_{:,j}^r$ jointly for all rows $i \in I_j \subseteq \{1, \dots, m\}$ conditional on the factors $\mathbf{f}_r = (\mathbf{f}_1^r, \dots, \mathbf{f}_T^r)$, the remaining indicators $\delta_{i,-j}^r$ and the hyperparameter τ_j , without conditioning on the model parameters β_r and $\sigma_1^2, \dots, \sigma_m^2$ using following steps:

- (I-a) Compute the log likelihood ratio O_{ij} for all rows $i \in I_j$ where all elements of $\delta_{i,-j}$ are zero as

$$O_{ij} = \log \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 1, \delta_{i,-j}, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 0, \delta_{i,-j}, \mathbf{f}_r)} = \log \frac{\Gamma(c_T)(C_{iT}^n)^{c_T}}{\Gamma(c_T^n)(C_{iT}^n)^{c_T}} + D_{ij}. \quad (\text{D.7})$$

c_T^n and C_{iT}^n are the posterior moments of the null model given in (C.7). c_T and C_{iT} are the posterior moments of σ_i^2 for a dedicated measurement with $j_i = j$, given in (C.12) and (C.13), respectively, for a hierarchical Gaussian and a fractional prior. For a hierarchical Gaussian prior, $D_{ij} = 0.5 \log(B_{iT}/B_{i0,jj})$, where $B_{i0,jj}$ is j th diagonal element of the prior variance $\mathbf{B}_{i0}$ and B_{iT} is the posterior scale factor for a dedicated measurement with $j_i = j$, given in (C.12). For a fractional prior, $D_{ij} = 0.5 \log(b(2\pi)^{bT})$.

This step is trivial to vectorize.

- (I-b) For all rows $i \in \{i_1, \dots, i_n\} \subseteq I_j$ where $\delta_{i,-j}$ is non-zero, compute

$$O_{ij} = \log \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 1, \delta_{i,-j}, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 0, \delta_{i,-j}, \mathbf{f}_r)} = c_T \log \frac{C_{iT}^0}{C_{iT}^1} + D_{ij}, \quad (\text{D.8})$$

where c_T and $C_{iT}^{\delta_{ij}}$ are the posterior moments of $\sigma_i^2 | \delta_{ij}, \cdot$ given in (C.8) and C_{iT}^0 refers to a model with $\delta_{ij} = 0$, while C_{iT}^1 refer to a model with $\delta_{ij} = 1$. For a hierarchical Gaussian prior,

$$D_{ij} = 0.5 \log(|\mathbf{B}_{iT}^1|/|\mathbf{B}_{iT}^0|) - 0.5 \log(|\mathbf{B}_{i0}^1|/|\mathbf{B}_{i0}^0|), \quad (\text{D.9})$$

where $\mathbf{B}_{i0}^{\delta_{ij}}$ and $\mathbf{B}_{iT}^{\delta_{ij}}$ refer to the prior and posterior moments of $\beta_i^{\delta_{ij}} | \delta_{ij}, \cdot$ given in (C.9). $\mathbf{B}_{i0}^1$ and $\mathbf{B}_{iT}^1$ refer to the prior and posterior moments for a model where $\delta_{ij} = 1$, while $\mathbf{B}_{i0}^0$ and $\mathbf{B}_{iT}^0$ refer to the prior and posterior moments for a model where $\delta_{ij} = 0$. For a fractional prior, $D_{ij} \equiv 0.5 \log b$.

Use Algorithm D.2 to determine C_{iT}^1 , C_{iT}^0 , as well as D_{ij} for a hierarchical Gaussian prior, simultaneously for all rows $i \in \{i_1, \dots, i_n\} \subseteq I_j$.

- (I-c) Determine the vector of the log posterior odds $O_{ij}^{\text{post}} = O_{ij} + O_{ij}^{\text{pr}}$ for all rows $i \in I_j$. Joint sampling of $\delta_{ij} | \tau_j, \cdot$ is easily vectorized:

- (I-c1) Propose $\delta_{ij}^{\text{new}} = 1 - \delta_{ij}$ for $i \in I_j$.
- (I-c2) Draw a vector of $|I_j|$ random variables $U_i \sim \mathcal{U}[0, 1]$, indexed by $i \in I_j$.
- (I-c3) For all rows $i \in I_j$, where $\delta_{ij} = 0$, accept the proposal $\delta_{ij}^{\text{new}} = 1$ iff $\log U_i \leq O_{ij}^{\text{post}}$;
- (I-c4) For all rows $i \in I_j$, where $\delta_{ij} = 1$, accept the proposal $\delta_{ij}^{\text{new}} = 0$ iff $\log U_i \leq -O_{ij}^{\text{post}}$.

Using, respectively, (D.2) and (D.3), the expression for O_{ij} in (D.8) is easily derived. Since the indicators in column j are independent given τ_j , Step (I-c) is based on $|I_j|$ independent Metropolis-Hastings (MH) steps each of which proposes to update δ_{ij} by flipping the indicator, i.e. $\delta_{ij}^{\text{new}} = 1 - \delta_{ij}$.⁷ It is easy to verify that the acceptance rules formulated in Step (I-c3) and (I-c4) are equivalent to the more conventional rule to accept δ_{ij}^{new} with probability

$$\min \left\{ 1, \frac{\Pr(\delta_{ij}^{\text{new}} | \boldsymbol{\delta}_{i,-j}, \tau_j, \tilde{\mathbf{y}}_i, \mathbf{f}_r)}{\Pr(\delta_{ij} | \boldsymbol{\delta}_{i,-j}, \tau_j, \tilde{\mathbf{y}}_i, \mathbf{f}_r)} \right\} = \min \{1, \exp(O_{ij}^{\text{post}})\}.$$

Algorithm D.2. To compute all relevant posterior moments in (D.8) simultaneously for all rows $\{i_1, \dots, i_n\}$, proceed as follows:

- (a) Set the indicator $\delta_{i_l,j} = 1$ in each row $i_l \in \{i_1, \dots, i_n\}$. Reorder the columns of the factor loading matrix in such a way, that the j th column appears last. This is simply done by permuting the column of $\mathbf{F}$ appropriately before defining $\mathbf{X}_{i_l}^\delta$.⁸
- (b) Set up the information matrix $\mathbf{P}$ and the covector $\mathbf{m}$ of the corresponding joint posterior of all non-zero factor loadings in the rows $i_1, \dots, i_n$ as described in Algorithm C.1. Compute the Cholesky decomposition $\mathbf{L}$ of $\mathbf{P}$ and the corresponding vector $\mathbf{x}$ solving $\mathbf{L}\mathbf{x} = \mathbf{m}$.
- (c) Knowing $\mathbf{L}$ and $\mathbf{x}$, a vectorized computation of the log likelihood ratio (D.7) for all rows $i_l \in \{i_1, \dots, i_n\}$ is possible. The posterior moments $C_{i_l,T}^1$ are directly available from the appropriate subvectors $\mathbf{x}_{i_l}$ of $\mathbf{x}$, defined in (C.14). When we switch from $\delta_{i_l,j} = 1$ to a model where $\delta_{i_l,j} = 0$, then for a hierarchical Gaussian prior,

$$C_{i_l,T}^0 = C_{i_l,T}^1 + \frac{1}{2}(x_{i_l}^*)^2, \quad (\text{D.10})$$

where $x_{i_l}^* = (\mathbf{x}_{i_l})_{q_{i_l}}$ is the last element of $\mathbf{x}_{i_l}$. Furthermore,

$$0.5 \log(|\mathbf{B}_{i_l,T}^1|/|\mathbf{B}_{i_l,T}^0|) = -\log L_{i_l}^*, \quad (\text{D.11})$$

⁷Alternatively, a Gibbs step may be used, i.e. set $\delta_{ij}^{\text{new}} = 1$ iff $\log(U_i/(1 - U_i)) \leq O_{ij}^{\text{post}}$, otherwise $\delta_{ij}^{\text{new}} = 0$. However, simulation experiments indicate that the MH step is more efficient.

⁸While the fractional prior is not affected by this, it might be necessary to reorder the prior mean and the prior covariance matrix for a hierarchical Gaussian prior.

where $L_{i_l}^* = (\mathbf{L}_i)_{q_{i_l}, q_{i_l}}$ is the last diagonal element of the submatrix $\mathbf{L}_{i_l}$. Therefore,

$$D_{ij} = -\log L_{i_l}^* - 0.5 \log B_{i0, jj}.$$

For the fractional prior,

$$C_{i_l, T}^0 = C_{i_l, T}^1 + \frac{1-b}{2}(x_{i_l}^*)^2, \quad (\text{D.12})$$

where $x_{i_l}^* = (\mathbf{x}_{i_l})_{q_{i_l}}$ is again the last element of $\mathbf{x}_{i_l}$.

Derivation of Step (c). When we switch from a model where all indicator $\delta_{i_1, j} = \dots = \delta_{i_n, j} = 1$ are equal to one to a model where all indicators $\delta_{i_1, j} = \dots = \delta_{i_n, j} = 0$ are zero, then the information matrix $\mathbf{P}^0$ and the covector $\mathbf{m}^0$ of the joint posterior of the remaining non-zero factor loadings is obtained from $\mathbf{P}$ and $\mathbf{m}$ simply by deleting all rows and columns corresponding to $\delta_{i_1, j}, \dots, \delta_{i_n, j}$, and the Cholesky decomposition $\mathbf{L}^0$ of $\mathbf{P}^0$ is obtained from $\mathbf{L}$ in the same way. Also the vector $\mathbf{x}^0$ solving $\mathbf{L}^0 \mathbf{x}^0 = \mathbf{m}^0$ is obtained from $\mathbf{x}$ simply by deleting the rows corresponding to $\delta_{i_1, j}, \dots, \delta_{i_n, j}$. This last result is easily seen by considering the subsystem $\mathbf{L}_{i_l} \mathbf{x}_{i_l} = \mathbf{m}_{i_l, T}^\delta$ corresponding to the i_l th row. Because

$$\mathbf{L}_{i_l} = \begin{pmatrix} \mathbf{L}_{i_l}^0 & \mathbf{O} \\ \mathbf{l}_{i_l} & L_{i_l}^* \end{pmatrix} = \begin{pmatrix} \mathbf{L}_{i_l}^0 & \mathbf{O} \\ \mathbf{l}_{i_l} & L_{i_l}^* \end{pmatrix}, \quad (\text{D.13})$$

we obtain $\mathbf{L}_{i_l}^0 \mathbf{x}_{i_l}^0 = \mathbf{m}_{i_l}^0$, where $\mathbf{x}_{i_l}^0$ is obtained from $\mathbf{x}_{i_l}$ by deleting the q_{i_l} th element $x_{i_l}^* = (\mathbf{x}_{i_l})_{q_{i_l}}$. Hence, $\mathbf{x}_{i_l}^0$ defines the desired subvector of $\mathbf{x}^0$ to compute $C_{i_l, T}^0$ as in (C.14). Since $(\mathbf{x}_{i_l}^0)' \mathbf{x}_{i_l}^0 = \mathbf{x}_{i_l}' \mathbf{x}_{i_l} - (\mathbf{x}_{i_l})_{q_{i_l}}^2$ we obtain from (C.8) that (D.12) holds. Note, however, that this simple relationship would not hold without reordering the columns as described above.

Finally, to compute the log likelihood ratio for a hierarchical Gaussian prior, the ratio of the determinants $|\mathbf{B}_{i_l, T}^1|/|\mathbf{B}_{i_l, T}^0|$ is required. Since the lower triangular matrices $\mathbf{L}_{i_l}^1$ and $\mathbf{L}_{i_l}^0$ are, respectively, the Cholesky decomposition of $(\mathbf{B}_{i_l, T}^1)^{-1}$ and $(\mathbf{B}_{i_l, T}^0)^{-1}$, we obtain:

$$1/|\mathbf{B}_{i_l, T}^1|^{1/2} = |(\mathbf{B}_{i_l, T}^1)^{-1}|^{1/2} = |\mathbf{L}_{i_l}^1|, \quad (\text{D.14})$$

where $|\mathbf{L}_{i_l}^1|$ is the product of the diagonal elements of $\mathbf{L}_{i_l}^1$. Computing $|\mathbf{B}_{i_l, T}^0|$ in the same way and using (D.13) proves (D.11).

Appendix E: Updating shrinkage parameters in Step (S)

Under a hierarchical Gaussian prior on the factor loadings β_{ij} , Steps (P), (D) and (L) of Algorithm 1 are performed conditional on all hyperparameters of this prior. Similarly, for the F-prior (3.15) on $\sigma_1^2, \dots, \sigma_m^2$, these steps are performed conditional on the scaling parameters $C_{10}, \dots, C_{m0}$. All unknown scaling parameters in the prior of the factor loadings β_{ij} and/or the prior of the idiosyncratic variances σ_i^2 are updated in Step (S)

of Algorithm 1. This step is based on [Knaus and Frühwirth-Schnatter \(2023\)](#) and relies on a representation of an F-distribution as a gamma scale mixture of inverse gamma distributions and avoids the GIG-distribution which was used in [Cadonna et al. \(2020\)](#).⁹

Step (S-a). Under prior (3.12), sample the local shrinkage parameters ω_{ij} using the representation of the F-distribution as a scale mixture of inverse gamma distributions:

$$\omega_{ij}|b_{ij}^\omega \sim \mathcal{G}^{-1}(c^\omega, b_{ij}^\omega), \quad b_{ij}^\omega \sim \mathcal{G}\left(a^\omega, \frac{a^\omega}{c^\omega}\right).$$

This yields a two-step sampler, where $b_{ij}^\omega|\omega_{ij}$ is imputed from

$$b_{ij}^\omega|\omega_{ij} \sim \mathcal{G}\left(a^\omega + c^\omega, \frac{a^\omega}{c^\omega} + \frac{1}{\omega_{ij}}\right), \quad (\text{E.1})$$

and $\omega_{ij}|b_{ij}^\omega, \beta_{ij}, \delta_{ij}, \kappa, \theta_j$ given b_{ij}^ω is sampled from

$$\omega_{ij}|b_{ij}^\omega, \beta_{ij}, \delta_{ij}, \kappa, \theta_j \sim \mathcal{G}^{-1}\left(c^\omega + \delta_{ij}\frac{1}{2}, b_{ij}^\omega + \delta_{ij}\frac{\beta_{ij}^2}{2\kappa\theta_j\sigma_i^2}\right). \quad (\text{E.2})$$

This has to be done for the entire matrix $\boldsymbol{\omega}_r = (\omega_{ij})$, as the local shrinkage parameters ω_{ij} are needed to compute the odds likelihood ratio O_{ij} of $\delta_{ij} = 1$ versus $\delta_{ij} = 0$ in Step (L) also above the current pivot, see Appendix G.

Step (S-b). Sample the column-specific shrinkage parameters $\boldsymbol{\theta}_r = (\theta_1, \dots, \theta_r)$, if any. For $\theta_j \sim \mathcal{G}^{-1}(c^\theta, b_j^\theta)$, this yields for all $j = 1$ to r :

$$\theta_j|\kappa, \beta_r, \boldsymbol{\Sigma}_r, \boldsymbol{\delta}_r, \boldsymbol{\omega}_r, b_j^\theta \sim \mathcal{G}^{-1}\left(c^\theta + \frac{d_j}{2}, b_j^\theta + \frac{1}{2\kappa} \sum_{i=1}^m \delta_{ij} \frac{\beta_{ij}^2}{\sigma_i^2 \omega_{ij}}\right). \quad (\text{E.3})$$

For $\theta_j \sim \text{F}(2a^\theta, 2c^\theta)$, we use again the representation of the F-distribution as a scale mixture of inverse gamma distributions:

$$\theta_j|b_j^\theta \sim \mathcal{G}^{-1}(c^\theta, b_j^\theta), \quad b_j^\theta \sim \mathcal{G}\left(a^\theta, \frac{a^\theta}{c^\theta}\right),$$

⁹A r.v. $X \sim \text{F}(2a, 2c)$ has a representation as a gamma scale mixture of inverse gamma distributions:

$$X|b \sim \mathcal{G}^{-1}(c, b), \quad b \sim \mathcal{G}\left(a, \frac{a}{c}\right).$$

Hence,

$$b|X \sim \mathcal{G}\left(a + c, \frac{a}{c} + \frac{1}{X}\right).$$

If $Y|X \sim \mathcal{N}(0, dX)$, then

$$X|b, Y \sim \mathcal{G}^{-1}\left(c + \frac{1}{2}, b + \frac{Y^2}{2d}\right).$$

and impute $b_j^\theta | \theta_j$ from

$$b_j^\theta | \theta_j \sim \mathcal{G} \left(a^\theta + c^\theta, \frac{a^\theta}{c^\theta} + \frac{1}{\theta_j} \right), \quad (\text{E.4})$$

before we update $\theta_j | \kappa, \beta_r, \Sigma_r, \delta_r, \omega_r, b_j^\theta$ from (E.3).

Step (S-c). If κ is random with prior $\kappa \sim \mathcal{G}^{-1}(c^\kappa, b^\kappa)$, then $\kappa | \beta_r, \Sigma_r, \omega_r, \delta_r, \theta_r$ is updated from

$$\kappa | \beta_r, \Sigma_r, \omega_r, \delta_r, \theta_r \sim \mathcal{G}^{-1} \left(c^\kappa + \frac{d}{2}, b^\kappa + 1/2 \sum_{i=1}^m \frac{1}{\sigma_i^2} \sum_{j=1}^r \delta_{ij} \frac{\beta_{ij}^2}{\theta_j \omega_{ij}} \right), \quad (\text{E.5})$$

where $d = \sum_{i=1}^m \sum_{j=1}^r \delta_{ij}$ is the total number of non-zero loadings in the CFA model. If $\kappa \sim \text{F}(2a^\kappa, 2c^\kappa)$, then using the same representation of the F-distribution as above, we impute $b^\kappa | \kappa$ from

$$b^\kappa | \kappa \sim \mathcal{G} \left(a^\kappa + c^\kappa, \frac{a^\kappa}{c^\kappa} + \frac{1}{\kappa} \right), \quad (\text{E.6})$$

before we update $\kappa | \beta_r, \Sigma_r, \omega_r$ from (E.5).

Step (S-d). For the F-prior (3.15) on $\sigma_1^2, \dots, \sigma_m^2$, the scaling parameters $C_{10}, \dots, C_{m0}$ are updated in the following way. Using that $p(C_{i0} | b_i^\sigma, \sigma_i^2)$ is proportional to

$$\begin{aligned} p(C_{i0} | b_i^\sigma, \sigma_i^2) &\propto p(\sigma_i^2 | C_{i0}, b_i^\sigma) p(C_{i0} | b_i^\sigma) \\ &\propto (C_{i0})^{c^\sigma} \exp \left\{ -\frac{C_{i0}}{\sigma_i^2} \right\} (C_{i0})^{a^\sigma - 1} \exp \left\{ -\frac{a^\sigma C_{i0}}{b_i^\sigma} \right\} \\ &\propto (C_{i0})^{a^\sigma + c^\sigma - 1} \exp \left\{ -C_{i0} \left(\frac{a^\sigma}{b_i^\sigma} + \frac{1}{\sigma_i^2} \right) \right\}, \end{aligned}$$

$C_{i0} | b_i^\sigma, \sigma_i^2$ is sampled from the corresponding gamma distribution:

$$C_{i0} | b_i^\sigma, \sigma_i^2 \sim \mathcal{G} \left(a^\sigma + c^\sigma, \frac{a^\sigma}{b_i^\sigma} + \frac{1}{\sigma_i^2} \right). \quad (\text{E.7})$$

Appendix F: Details on split and merge moves in Step (R)

Proof of (4.3). For any zero column j_{sp} in a sparsity matrix δ_H with r active and r_{sp} spurious columns, there are $m - (r + r_{\text{sp}})$ unrestricted elements that can be subjected to variable selection and take the value one with probability $\tau_{j_{\text{sp}}} | H \sim \mathcal{B}(a_H, b_H)$ following the prior (3.2). Hence,

$$\Pr(d_{j_{\text{sp}}} = 0 | r, r_{\text{sp}}) = \int (1 - \tau_{j_{\text{sp}}})^{m - (r + r_{\text{sp}})} p(\tau_{j_{\text{sp}}}) d\tau_{j_{\text{sp}}} =$$

$$\frac{B(a_H, b_H + m - (r + r_{sp}))}{B(a_H, b_H)}, \quad (\text{F.1})$$

whereas

$$\begin{aligned} \Pr(d_{j_{sp}} = 1 | r, r_{sp}) &= (m - (r + r_{sp})) \int \tau_{j_{sp}} (1 - \tau_{j_{sp}})^{m - (r + r_{sp}) - 1} p(\tau_{j_{sp}}) d\tau_{j_{sp}} = \\ &= \frac{(m - (r + r_{sp})) B(a_H + 1, b_H + m - (r + r_{sp}) - 1)}{B(a_H, b_H)}. \end{aligned} \quad (\text{F.2})$$

Therefore,

$$\frac{\Pr(d_{j_{sp}} = 1 | r, r_{sp})}{\Pr(d_{j_{sp}} = 0 | r, r_{sp})} = \frac{a_H(m - r - r_{sp})}{b_H + m - r - r_{sp} - 1}.$$

Proposing factors in a spurious column - Proof of (4.1). Whenever a zero column is turned into a spurious column (with column index j_{sp}), factors $\mathbf{f}_{j_{sp}} = (f_{j_{sp},1}, \dots, f_{j_{sp},T})$ are proposed, while holding the factors $\mathbf{f}_t^r, t = 1, \dots, T$, in all active columns fixed. Draws of $f_{j_{sp},t}$ could be proposed from the prior $f_{j_{sp},t} \sim \mathcal{N}(0, 1)$, since column j_{sp} was a zero column before splitting. However, with l_{sp} being the pivot row, $y_{l_{sp},t}$ is a measurement that contains information about $f_{j_{sp},t}$ in a spurious column and its likelihood can be combined with the prior to define the conditional posterior density $p(f_{j_{sp},t} | \mathbf{f}_t^r, \beta_r, \sigma_{l_{sp}}^2, y_{l_{sp},t})$ given $y_{l_{sp},t}$ in addition to $\mathbf{f}_t^r, \beta_r$ and $\sigma_{l_{sp}}^2$.

It is easy to verify from the filter given in (4.2) that for a spurious column j_{sp} with leading element $\Xi_{l_{sp}}$, the conditional density $p(f_{j_{sp},t} | \mathbf{f}_t^r, \beta_r, \sigma_{l_{sp}}^2, y_{l_{sp},t})$ of $f_{j_{sp},t}$ is given by:

$$\begin{aligned} f_{j_{sp},t} | \mathbf{f}_t^r, \beta_r, \sigma_{l_{sp}}^2, y_{l_{sp},t} &\sim \mathcal{N}(E_{j_{sp},t}, V_{j_{sp}}), \\ V_{j_{sp}} &= \left(1 + \frac{\Xi_{l_{sp}}^2}{\sigma_{l_{sp}}^{2,\text{new}}}\right)^{-1} = \frac{\sigma_{l_{sp}}^{2,\text{new}}}{\sigma_{l_{sp}}^{2,\text{new}} + \Xi_{l_{sp}}^2}, \\ E_{j_{sp},t} &= \frac{V_{j_{sp}} \Xi_{l_{sp}}}{\sigma_{l_{sp}}^{2,\text{new}}} \tilde{y}_{l_{sp},t} = \frac{\Xi_{l_{sp}}}{\sigma_{l_{sp}}^{2,\text{new}} + \Xi_{l_{sp}}^2} \tilde{y}_{l_{sp},t}, \end{aligned}$$

where the pseudo outcome $\tilde{y}_{l_{sp},t}$ is given by $\tilde{y}_{l_{sp},t} = y_{l_{sp},t} - \beta_{r,l_{sp}} \mathbf{f}_t^r$. Using the definition of $\Xi_{l_{sp}}$ and $\sigma_{l_{sp}}^{2,\text{new}}$ in terms of $\sigma_{l_{sp}}^2$ and a uniform random variable $U_{j_{sp}} \in [-1, 1]$ given in (4.6), we obtain the simple expressions for the posterior moments given in (4.1):

$$V_{j_{sp}} = \frac{\sigma_{l_{sp}}^{2,\text{new}}}{\sigma_{l_{sp}}^{2,\text{new}} + \Xi_{l_{sp}}^2} = 1 - U_{j_{sp}}^2, \quad E_{j_{sp},t} = \frac{\Xi_{l_{sp}}}{\sigma_{l_{sp}}^{2,\text{new}} + \Xi_{l_{sp}}^2} \tilde{y}_{l_{sp},t} = \frac{U_{j_{sp}}}{\sqrt{\sigma_{l_{sp}}^2}} \tilde{y}_{l_{sp},t}.$$

Turning spurious into active columns. In Step (R-D) of Algorithm 1, variable selection is performed on the spurious columns of the current EFA model by marginalizing over the idiosyncratic variances and the factor loading matrix (β_r, Ξ) as in Step (D). By

starting with the last spurious column, we ensure that the application of Step (D) is valid. Indeed, ordering the spurious pivots in Step (L) by size guarantees that all rows of the current EFA model below l_{sp} are equal to the corresponding rows of the current CFA model in the first r columns and zero in all other columns. Hence, updating the zeros below l_{sp} can be regarded as an attempt to introduce an additional active column in the current CFA model and to increase r , while r_{sp} decreases. If a column remains spurious, then we integrate in the current EFA model immediately over the corresponding factor, while the current number of spurious columns r_{sp} is preserved.

Step (R-D) for hierarchical Gaussian priors with unknown shrinkage factors. Step (R-D) has to be adjusted slightly for hierarchical Gaussian priors with unknown shrinkage factors. A column-specific shrinkage parameter $\theta_{j_{\text{sp}}}$ is needed to compute the odds of turning zero loadings in the spurious column into non-zero ones. In addition, under the shrinkage prior (3.12), local shrinkage parameters $\omega_{i,j_{\text{sp}}}$ are needed for all rows $i = 1, \dots, m$ in the spurious column j_{sp} (note that $\omega_{i,j_{\text{sp}}}$ will be needed later in step (L), even if $i < l_{\text{sp}}$ lies above the spurious pivot element). All these quantities could be simply simulated from the prior. However, we found considerably gain in exploited the information contained in $\Xi_{l_{\text{sp}}}$ and $\sigma_{l_{\text{sp}}}^{2,\text{new}}$, when sampling $\theta_{j_{\text{sp}}}$ and $\omega_{l_{\text{sp}},j_{\text{sp}}}$, which basically results from conditioning on the r.v. $U_{j_{\text{sp}}}$.

For prior (3.11), the information contained in $\Xi_{l_{\text{sp}}}$ and $\sigma_{l_{\text{sp}}}^{2,\text{new}}$ is exploited by sampling $\theta_{j_{\text{sp}}} | \Xi_{l_{\text{sp}}}, \sigma_{l_{\text{sp}}}^{2,\text{new}}$ from (E.3), using that $\omega_{ij} = 1$:

$$\theta_{j_{\text{sp}}} | \kappa, U_{j_{\text{sp}}} \sim \mathcal{G}^{-1} \left(c^\theta + \frac{1}{2}, b_{j_{\text{sp}}}^\theta + \frac{U_{j_{\text{sp}}}^2}{2\kappa(1 - U_{j_{\text{sp}}}^2)} \right), \quad (\text{F.3})$$

where the scale parameter is obtained from combining (4.6) and (E.3):

$$\frac{\beta_{l_{\text{sp}},j_{\text{sp}}}^2}{(\Sigma_{r+1})_{l_{\text{sp}},l_{\text{sp}}}} = \frac{\Xi_{l_{\text{sp}}}^2}{\sigma_{l_{\text{sp}}}^{2,\text{new}}} = \frac{U_{j_{\text{sp}}}^2}{1 - U_{j_{\text{sp}}}^2}.$$

Under the prior $\theta_{j_{\text{sp}}} \sim \mathcal{G}^{-1}(c^\theta, b^\theta)$, the scale $b_{j_{\text{sp}}}^\theta = b^\theta$, whereas under the F-prior on $\theta_{j_{\text{sp}}}$, $b_{j_{\text{sp}}}^\theta$ is imputed from the prior. This is achieved by sampling $\theta_{j_{\text{sp}}} \sim \text{F}(2a^\theta, 2c^\theta)$ and $b_{j_{\text{sp}}}^\theta | \theta_{j_{\text{sp}}}$ from (E.4):

$$b_{j_{\text{sp}}}^\theta | \theta_{j_{\text{sp}}} \sim \mathcal{G} \left(a^\theta + c^\theta, \frac{a^\theta}{c^\theta} + \frac{1}{\theta_{j_{\text{sp}}}} \right).$$

For the shrinkage prior (3.12), first local shrinkage parameters $\omega_{i,j_{\text{sp}}}$ are sampled from the prior for all rows, i.e. $\omega_{i,j_{\text{sp}}} \sim \text{F}(2a^\omega, 2c^\omega)$, $i = 1, \dots, m$. The information contained in $\Xi_{l_{\text{sp}}}$, $\sigma_{l_{\text{sp}}}^{2,\text{new}}$ and $\omega_{l_{\text{sp}},j_{\text{sp}}}$ is then exploited by sampling $\theta_{j_{\text{sp}}} | \Xi_{l_{\text{sp}}}, \sigma_{l_{\text{sp}}}^{2,\text{new}}, \omega_{l_{\text{sp}},j_{\text{sp}}}$ from (E.3):

$$\theta_{j_{\text{sp}}} | \kappa, U_{j_{\text{sp}}}, \omega_{l_{\text{sp}},j_{\text{sp}}} \sim \mathcal{G}^{-1} \left(c^\theta + \frac{1}{2}, b_{j_{\text{sp}}}^\theta + \frac{U_{j_{\text{sp}}}^2}{2\kappa\omega_{l_{\text{sp}},j_{\text{sp}}}(1 - U_{j_{\text{sp}}}^2)} \right),$$

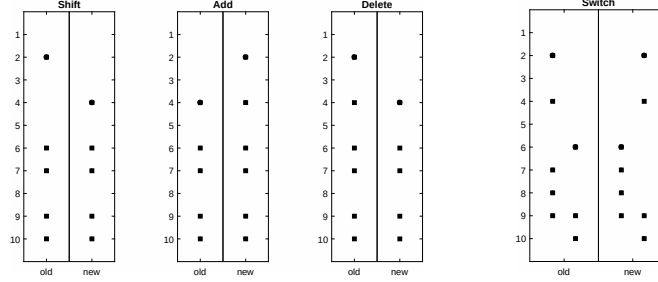


Figure G.1: MH moves to change the pivot rows of an unordered GLT structure; from left to right: shifting the pivot row, adding a new pivot row, deleting a pivot row and switching the pivot rows (and some indicators below) in two columns.

where $b_{j_{\text{sp}}}^\theta$ is defined as above for prior (3.11). Finally, we resample $\omega_{l_{\text{sp}}, j_{\text{sp}}}$ by first imputing $b_{l_{\text{sp}}, j_{\text{sp}}}^\omega$ from the prior by sampling $b_{l_{\text{sp}}, j_{\text{sp}}}^\omega | \omega_{l_{\text{sp}}, j_{\text{sp}}}$ from (E.1):

$$b_{l_{\text{sp}}, j_{\text{sp}}}^\omega | \omega_{l_{\text{sp}}, j_{\text{sp}}} \sim \mathcal{G} \left(a^\omega + c^\omega, \frac{a^\omega}{c^\omega} + \frac{1}{\omega_{l_{\text{sp}}, j_{\text{sp}}}} \right),$$

(where $\omega_{l_{\text{sp}}, j_{\text{sp}}}$ has been sampled from the prior in the first step) and then sampling from (E.2):

$$\omega_{l_{\text{sp}}, j_{\text{sp}}} | b_{l_{\text{sp}}, j_{\text{sp}}}^\omega, \theta_{j_{\text{sp}}}, \kappa, U_{j_{\text{sp}}} \sim \mathcal{G}^{-1} \left(c^\omega + \frac{1}{2}, b_{l_{\text{sp}}, j_{\text{sp}}}^\omega + \frac{U_{j_{\text{sp}}}^2}{2\kappa\theta_{j_{\text{sp}}}(1 - U_{j_{\text{sp}}}^2)} \right).$$

In this way, we exploit the information contained in $\Xi_{l_{\text{sp}}}$ and $\sigma_{l_{\text{sp}}}^{2, \text{new}}$ for both shrinkage parameters.

Appendix G: Step (L) - Updating the pivots

This subsection provides details on Step (L) of Algorithm 1, which was briefly discussed in Section 4.2. Four local moves are applied which are illustrated in Figure G.1.

Shifting the pivot. A shift move is selected with probability p_{shift} . Let $l_\star$ denote the index of the first non-zero row below l_j (i.e. $\delta_{l_\star, j} = 1, \delta_{ij} = 0, l_j < i < l_\star$). If $l_\star > 2$, then a new pivot l_j^{new} is proposed by sampling uniformly from the set $\mathcal{M}(l_\star, \mathbf{l}_{r, -j}) = \{1, \dots, l_\star - 1\} \cap \mathcal{L}(\mathbf{l}_{r, -j})$, where $\mathcal{L}(\mathbf{l}_{r, -j})$ is the set of pivots outside of column j . If $\mathcal{M}(l_\star, \mathbf{l}_{r, -j})$ is empty, then no shift move is performed. Otherwise, two indicators in column j are changed, namely $\delta_{l_j^{\text{new}}, j}$ from zero to one and $\delta_{l_j, j}^{\text{new}}$ from one to zero, while the remaining elements of $\boldsymbol{\delta}_r$ are unchanged. The new indicator matrix $\boldsymbol{\delta}_r^{\text{new}}$ is accepted with probability $\min(1, \alpha_{\text{shift}})$, where

$$\alpha_{\text{shift}} = \exp(O_{l_j^{\text{new}}, j} - O_{l_j, j}) R_{\text{shift}},$$

with O_{ij} being the log likelihood ratio of $\delta_{ij} = 1$ versus $\delta_{ij} = 0$ defined in (D.6). A shift move does not change the number of non-zero elements d_j in column j and the prior ratio (4.7) of this move simplifies to:

$$R_{\text{shift}} = \frac{B(a_H + d_j - 1, b_H + m - l_j^{\text{new}} - d_j + 1)}{B(a_H + d_j - 1, b_H + m - l_j - d_j + 1)}. \quad (\text{G.1})$$

Since l_j^{new} is sampled from a set $\mathcal{M}(l_\star, \mathbf{l}_{r,-j})$ that does not depend on l_j , the proposal density is symmetric and the proposal ratio $q(l_j^{\text{new}} | l_\star, \mathbf{l}_{r,-j}) / q(l_j | l_\star, \mathbf{l}_{r,-j})$ cancels from α_{shift} .

Switching pivots. This move is selected for each column j with probability p_{switch} and is performed only if $r > 1$. A non-zero column $\ell \neq j$ is selected randomly and all indicators between (and including) row l_j and l_ℓ that are different are switched between the two columns, i.e. $\delta_{ij}^{\text{new}} = 1 - \delta_{ij}$ and $\delta_{i\ell}^{\text{new}} = 1 - \delta_{i\ell}$ for all $i \in \mathcal{S}_{j,\ell} = \{i : \min(l_\ell, l_j) \leq i \leq \max(l_\ell, l_j), \delta_{ij} \neq \delta_{i\ell}\}$. Evidently, this move switches pivots between the two columns. Since the corresponding proposal density satisfies $q(\boldsymbol{\delta}_r^{\text{new}} | \boldsymbol{\delta}_r) = q(\boldsymbol{\delta}_r | \boldsymbol{\delta}_r^{\text{new}})$, $\boldsymbol{\delta}_r^{\text{new}}$ is accepted with probability $\min(1, \alpha_{\text{switch}})$, where

$$\alpha_{\text{switch}} = \prod_{i \in \mathcal{S}_{j,\ell}} \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij}^{\text{new}}, \delta_{i\ell}^{\text{new}}, \boldsymbol{\delta}_r^u, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij}, \delta_{i\ell}, \boldsymbol{\delta}_r^u, \mathbf{f}_r)} R_{\text{switch}},$$

and $\boldsymbol{\delta}_r^u$ denotes all indicators that remain unchanged. The prior odds ratio of this move is derived from (4.7):

$$R_{\text{switch}} = \prod_{\tilde{l}=j,\ell} \frac{B(a_H + d_{\tilde{l}}^{\text{new}} - 1, b_H + m - l_{\tilde{l}}^{\text{new}} - d_{\tilde{l}}^{\text{new}} + 1)}{B(a_H + d_{\tilde{l}} - 1, b_H + m - l_{\tilde{l}} - d_{\tilde{l}} + 1)}. \quad (\text{G.2})$$

If $\delta_{ij} = 0$ (and consequently $\delta_{i\ell} = 1$), then we obtain:

$$\begin{aligned} & \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij}^{\text{new}} = 1, \delta_{i\ell}^{\text{new}} = 0, \boldsymbol{\delta}_r^u, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 0, \delta_{i\ell} = 1, \boldsymbol{\delta}_r^u, \mathbf{f}_r)} = \\ & \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij}^{\text{new}} = 1, \delta_{i\ell}^{\text{new}} = 0, \boldsymbol{\delta}_r^u, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 0, \delta_{i\ell}^{\text{new}} = 0, \boldsymbol{\delta}_r^u, \mathbf{f}_r)} \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 0, \delta_{i\ell}^{\text{new}} = 0, \boldsymbol{\delta}_r^u, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 0, \delta_{i\ell} = 1, \boldsymbol{\delta}_r^u, \mathbf{f}_r)} = \exp(O_{ij|\ell} - O_{i\ell|j}), \end{aligned}$$

where $O_{i,j_1|j_2}$ is the log likelihood ratio of $\delta_{i,j_1} = 1$ versus $\delta_{i,j_1} = 0$ provided that $\delta_{i,j_2} = 0$. It is obtained as the likelihood ratio O_{i,j_1} given in (D.6), with $\delta_{i,j_2} = 0$ for both models. On the other hand, if $\delta_{ij} = 1$ (and consequently $\delta_{i\ell} = 0$), then

$$\begin{aligned} & \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij}^{\text{new}} = 0, \delta_{i\ell}^{\text{new}} = 1, \boldsymbol{\delta}_r^u, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 1, \delta_{i\ell} = 0, \boldsymbol{\delta}_r^u, \mathbf{f}_r)} = \\ & \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij}^{\text{new}} = 0, \delta_{i\ell}^{\text{new}} = 1, \boldsymbol{\delta}_r^u, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij}^{\text{new}} = 0, \delta_{i\ell} = 0, \boldsymbol{\delta}_r^u, \mathbf{f}_r)} \frac{p(\tilde{\mathbf{y}}_i | \delta_{ij}^{\text{new}} = 0, \delta_{i\ell} = 0, \boldsymbol{\delta}_r^u, \mathbf{f}_r)}{p(\tilde{\mathbf{y}}_i | \delta_{ij} = 1, \delta_{i\ell} = 0, \boldsymbol{\delta}_r^u, \mathbf{f}_r)} = \exp(O_{i\ell|j} - O_{ij|\ell}). \end{aligned}$$

Therefore

$$\alpha_{\text{switch}} = R_{\text{switch}} \exp \left(\sum_{i \in \mathcal{S}_{j,\ell} : \delta_{ij}=0} (O_{ij|\ell} - O_{i\ell|j}) + \sum_{i \in \mathcal{S}_{j,\ell} : \delta_{ij}=1} (O_{i\ell|j} - O_{ij|\ell}) \right). \quad (\text{G.3})$$

This move allows changes in d_j and d_ℓ , but leaves the overall number d of non-zero elements unchanged.

Adding or deleting a pivot. Finally, a reversible pair of moves is selected with probability $1 - p_{\text{shift}} - p_{\text{switch}}$. The add move introduces a new pivot l_j^{new} in a row above the current pivot l_j which is not occupied by the pivots of the other columns. Hence, l_j^{new} is selected randomly from the set $\mathcal{A}(l_j, \mathbf{l}_{r,-j}) = \{1, \dots, l_j - 1\} \cap \mathcal{L}(\mathbf{l}_{r,-j})$, i.e. $\delta_{l_j^{\text{new}}, j}^{\text{new}} = 1$, while the remaining elements of $\boldsymbol{\delta}_r$ are unchanged (in particular $\delta_{l_j, j}^{\text{new}} = \delta_{l_j, j} = 1$). An add move is only possible if $|\mathcal{A}(l_j, \mathbf{l}_{r,-j})| > 0$.¹⁰

The corresponding reverse move is deterministic and deletes the current pivot l_j , turning $l_j^{\text{new}} = l_\star$ into the new pivot where $l_\star$ is the row index of the first non-zero element in $\boldsymbol{\delta}_r$ below l_j . Hence, $\delta_{l_j, j}^{\text{new}} = 0$, while all other elements of $\boldsymbol{\delta}_r$ remain unchanged. A delete move is only performed, if $l_\star$ is not a pivot in any other column of $\boldsymbol{\delta}_r$ (that is $l_\star \in \mathcal{L}(\mathbf{l}_{r,-j})$).

If neither an add nor a delete move is possible, then l_j remains unchanged. Otherwise, either an add or a delete move is selected with probability $p_{\text{add}}(\boldsymbol{\delta}_r)$ and $1 - p_{\text{add}}(\boldsymbol{\delta}_r)$. If both add and delete moves are possible, then $p_{\text{add}}(\boldsymbol{\delta}_r) = p_a$, with p_a being a tuning parameter; if only an add move is possible, then $p_{\text{add}}(\boldsymbol{\delta}_r) = 1$, whereas $p_{\text{add}}(\boldsymbol{\delta}_r) = 0$ if only a delete move is possible. Note that whenever an add move is selected, the reverse delete move is always possible; and similarly, the reverse add move is always possible, whenever a delete move is selected. This move changes d_j and increases or decreases the overall number of non-zero elements by one.

The acceptance probability for an add move is equal to $\min(1, \alpha_{\text{add}})$, with

$$\alpha_{\text{add}} = \exp(O_{l_j^{\text{new}}, j}) R_{\text{add}} \frac{|\mathcal{A}(l_j, \mathbf{l}_{r,-j})|(1 - p_{\text{add}}(\boldsymbol{\delta}_r^{\text{new}}))}{p_{\text{add}}(\boldsymbol{\delta}_r)},$$

where O_{ij} is the log likelihood ratio of $\delta_{ij} = 1$ versus $\delta_{ij} = 0$ defined in (D.6). Since $d_j^{\text{new}} = d_j + 1$, the prior ratio (4.7) of this move simplifies to:

$$R_{\text{add}} = \frac{B(a_H + d_j, b_H + m - l_j^{\text{new}} - d_j)}{B(a_H + d_j - 1, b_H + m - l_j - d_j + 1)}. \quad (\text{G.4})$$

The acceptance probability for a delete move is equal to $\min(1, \alpha_{\text{del}})$, with

$$\alpha_{\text{del}} = \exp(-O_{l_j, j}) R_{\text{del}} \frac{p_{\text{add}}(\boldsymbol{\delta}_r^{\text{new}})}{|\mathcal{A}(l_j^{\text{new}}, \mathbf{l}_{r,-j})|(1 - p_{\text{add}}(\boldsymbol{\delta}_r))}.$$

Since $d_j^{\text{new}} = d_j - 1$, the prior ratio (4.7) of this move simplifies to:

$$R_{\text{del}} = \frac{B(a_H + d_j - 2, b_H + m - l_j^{\text{new}} - d_j + 2)}{B(a_H + d_j - 1, b_H + m - l_j - d_j + 1)}. \quad (\text{G.5})$$

¹⁰The number of rows in $\mathcal{A}(l_j, \mathbf{l}_{r,-j})$ is equal to $l_j - z_j$, where $z_j = \#\{l_{j'} \in \mathbf{l}_r : l_{j'} < l_j\}$ is the rank of l_j among the pivots. Hence, an add move is possible, whenever $l_j > z_j$.

Tuning parameters. These four moves involve three tuning probabilities, namely p_{shift} , p_{switch} , and p_a , with $1 - p_{\text{shift}} - p_{\text{switch}} > 0$ and $0 < p_a < 1$.

Appendix H: More about boosting in Step (A)

Algorithm H.1 described a generic boosting strategy in Step (A) in Algorithm 1.

Algorithm H.1. Boosting MCMC. Step (A) in Algorithm 1 is implemented in three steps:

- (A-a) Choose a (current) value Ψ and move from the CFA model (2.3) to the expanded model (4.8) using the following transformation:

$$\tilde{\mathbf{f}}_t^r = (\Psi)^{1/2} \mathbf{f}_t^r, \quad \tilde{\beta}_r = \beta_r (\Psi)^{-1/2}. \quad (\text{H.1})$$

- (A-b) Sample a new value Ψ^{new} in model (4.8) conditional on $\tilde{\mathbf{f}}_r = (\tilde{\mathbf{f}}_1^r, \dots, \tilde{\mathbf{f}}_T^r)$ and $\tilde{\beta}_r$ from the conditional posterior $p(\Psi | \tilde{\mathbf{f}}_r, \tilde{\beta}_r)$ given by

$$p(\Psi | \tilde{\mathbf{f}}_r, \tilde{\beta}_r) \propto p(\Psi) p(\tilde{\beta}_r | \Psi) \prod_{j=1}^r \Psi_j^{-T/2} \exp \left\{ -\frac{1}{2\Psi_j} \sum_{t=1}^T \tilde{f}_{jt}^2 \right\}. \quad (\text{H.2})$$

- (A-c) Move back from model (4.8) to the CFA model (2.3) using the inverse of transformation (H.1) for $j = 1, \dots, r$:

$$\beta_{ij}^{\text{new}} = \beta_{ij} \sqrt{\Psi_j^{\text{new}} / \Psi_j}, \quad i = 1, \dots, m, \quad f_{jt}^{\text{new}} = f_{jt} \sqrt{\Psi_j / \Psi_j^{\text{new}}}, \quad t = 1, \dots, T.$$

Boosting affects the factor loading matrix β_r and all factors $\mathbf{f}_r = (\mathbf{f}_1^r, \dots, \mathbf{f}_T^r)$, see Step (A-c). The main difference between ASIS and MDA lies in the choice of the current value of Ψ in Step (A-a). While Ψ_j is chosen in a deterministic fashion for ASIS, typically involving a specific factor loading $\beta_{n_j,j}$ in each column, Ψ_j is sampled from a working prior for MDA. This leads to different priors $p(\Psi)$ and $p(\tilde{\beta}_r | \Psi)$ in (H.2).

ASIS for fractional priors. For boosting based on ASIS, a non-zero factor loading $\beta_{n_j,j}$ is chosen in column j to define the current value of Ψ_j as $\sqrt{\Psi_j} = \beta_{n_j,j}$. This creates a factor loading matrix $\tilde{\beta}_r$ in the expanded model (4.8) where $\tilde{\beta}_{n_j,j} = 1$ whereas $\tilde{\beta}_{i,j} = \beta_{ij} / \beta_{n_j,j}$ for $i \neq n_j$. $\beta_{n_j,j}$ can be chosen as the pivot element in each column, i.e. $n_j = l_j$, or such that $|\beta_{n_j,j}|$ is maximized for all loadings in column j . Apart from this choice, ASIS requires no further tuning. The implied prior of $\Psi_j = \beta_{n_j,j}^2$ is given by $p(\Psi_j) \propto \Psi_j^{-1/2}$, whereas $p(\tilde{\beta}_r | \Psi) \propto \prod_{j=1}^r \Psi_j^{(d_j-1)/2}$, since $p(\tilde{\beta}_{ij} | \sigma_i^2, \Psi_j, \delta_{ij} = 1) \propto \Psi_j^{1/2}$ for all $i \neq n_j$ with $\delta_{ij} = 1$. Hence, $p(\Psi | \tilde{\mathbf{f}}_r, \tilde{\beta}_r)$ defined in (H.2) reduces to a product of inverse gamma distributions where for all $j = 1, \dots, r$:

$$\Psi_j^{\text{new}} | \mathbf{f}_r, \beta_r, \delta_r \sim \mathcal{G}^{-1} \left(\frac{T - d_j}{2}, \frac{\beta_{n_j,j}^2}{2} \sum_{t=1}^T f_{jt}^2 \right),$$

where $d_j = \sum_{i=1}^m \delta_{ij}$ counts the non-zero elements in column j (which is at least equal to 2 by definition of δ_r).

MDA for fractional priors. In MDA, the current value Ψ is drawn from a working prior $p(\Psi)$ which is independent both of β_r and Σ_r . This guarantees that the prior distribution of β_r remains unchanged, despite moving between the two models. Below, an inverse gamma working prior $\Psi_j \sim \mathcal{G}^{-1}(\nu_j, q_j)$ is applied. The prior in the expanded model reads $p(\beta_r | \Psi) \propto \prod_{j=1}^r \Psi_j^{d_j/2}$, since $p(\beta_{ij} | \sigma_i^2, \Psi_j, \delta_{ij} = 1) \propto \Psi_j^{1/2}$ for all i with $\delta_{ij} = 1$. The two last terms in (H.2) factor into a product of independent inverse gamma distributions which leads to an inverse gamma posterior for each Ψ_j :

$$\Psi_j^{\text{new}} | \Psi_j, \mathbf{f}_r, \delta_r \sim \mathcal{G}^{-1} \left(\nu_j - d_j/2 + \frac{T}{2}, q_j + \Psi_j/2 \sum_{t=1}^T f_{tj}^2 \right).$$

Boosting for hierarchical Gaussian priors. Column boosting is based on interweaving θ_j into the state equation by choosing $\Psi_j = \theta_j$. This leads to a prior for the loadings $\tilde{\beta}_{ij}$ which is independent of θ_j , while θ_j acts as variance of the j th factors in the expanded model, where $\tilde{f}_{jt} | \theta_j \sim \mathcal{N}(0, \theta_j)$. Hence, in Step (A), θ_j is updated in the expanded model under the inverse gamma prior $\theta_j \sim \mathcal{G}^{-1}(c^\theta, b_j^\theta)$:

$$\theta_j^{\text{new}} | \theta_j, \mathbf{f}_r \sim \mathcal{G}^{-1} \left(c^\theta + \frac{T}{2}, B_j \right), \quad B_j = b_j^\theta + 1/2 \sum_{t=1}^T \tilde{f}_{jt}^2 = b_j^\theta + \theta_j/2 \sum_{t=1}^T f_{tj}^2.$$

Moving back, all factors and factor loadings in the CFA model are updated:

$$\beta_{ij}^{\text{new}} = \sqrt{\theta_j^{\text{new}}} \tilde{\beta}_{ij} = \sqrt{\frac{\theta_j^{\text{new}}}{\theta_j}} \beta_{ij}, \quad f_{jt}^{\text{new}} = \tilde{f}_{jt} / \sqrt{\theta_j^{\text{new}}} = \sqrt{\frac{\theta_j}{\theta_j^{\text{new}}}} f_{jt}.$$

A boosting step that we found to be useful for hierarchical Gaussian priors in addition to column boosting is interweaving the global shrinkage parameter κ into the prior $\theta_j \sim \mathcal{G}^{-1}(c^\theta, b_j^\theta)$ of the shrinkage parameter θ_j through the transformation $\tilde{\theta}_j = \kappa \theta_j$:

$$\begin{aligned} \beta_{ij} | \delta_{ij} = 1, \omega_{ij}, \tilde{\theta}_j, \sigma_i^2 &\sim \mathcal{N} \left(0, \tilde{\theta}_j \omega_{ij} \sigma_i^2 \right), \\ \tilde{\theta}_j | \kappa &\sim \mathcal{G}^{-1}(c^\theta, \kappa b_j^\theta), \quad \kappa \sim \mathcal{G}^{-1}(c^\kappa, b^\kappa). \end{aligned}$$

The global shrinkage parameter κ is then updated in Step (A) under this representation, combining the likelihood

$$p(\tilde{\theta}_1, \dots, \tilde{\theta}_r | \kappa) \propto \kappa^{rc^\theta} \exp \left(-\kappa \sum_{j=1}^r \frac{b_j^\theta}{\tilde{\theta}_j} \right),$$

which is the kernel of a gamma density in κ with the inverse gamma prior $p(\kappa)$. This yields a generalized inverse Gaussian distribution¹¹ to update κ^{new} :

$$\kappa^{\text{new}} | \theta_1, \dots, \theta_r, \kappa \sim \mathcal{GIG} \left(rc^\theta - c^\kappa, \frac{2}{\kappa} \sum_{j=1}^r \frac{b_j^\theta}{\theta_j}, 2b^\kappa \right).$$

Given κ^{new} , the column-specific parameters are updated as

$$\theta_j^{\text{new}} = \tilde{\theta}_j / \kappa^{\text{new}} = \frac{\kappa}{\kappa^{\text{new}}} \theta_j, \quad j = 1, \dots, r.$$

Row boosting for the F-prior (3.15) on $\sigma_1^2, \dots, \sigma_m^2$. Under the F-prior (3.15) on $\sigma_1^2, \dots, \sigma_m^2$, row boosting is applied in a CFA model with a loading matrix β_r of r active columns with elements β_{ij} and corresponding sparsity matrix δ_r with elements δ_{ij} . In the standard parametrization of the CFA model, each row is written as

$$y_{it} = \beta_{r,i} \mathbf{f}_t^r + \epsilon_{it}, \quad \epsilon_{it} | \sigma_i^2 \sim \mathcal{N}(0, \sigma_i^2), \quad (\text{H.3})$$

where $\beta_{r,i}$ is the i th row of β_r and $\sigma_i^2 | C_{i0} \sim \mathcal{G}^{-1}(c^\sigma, C_{i0})$ is “centered” around the random scale $C_{i0} \sim \mathcal{G}(a^\sigma, a^\sigma / b_i^\sigma)$. Row boosting is based on moving to a parametrization, where the random scale C_{i0} is moved from the prior $p(\sigma_i^2 | C_{i0})$ into the observation equation,

$$y_{it} = \beta_{r,i} \mathbf{f}_t^r + \epsilon_{it}, \quad \epsilon_{it} | C_{i0}, \tilde{\sigma}_i^2 \sim \mathcal{N}(0, C_{i0} \tilde{\sigma}_i^2), \quad (\text{H.4})$$

and the idiosyncratic variance $\tilde{\sigma}_i^2$ is non-centered, i.e. $\tilde{\sigma}_i^2 \sim \mathcal{G}^{-1}(c^\sigma, 1)$. ASIS can be applied to move between the two models for each row i to resample $(C_{i0})^{\text{new}}$ and define a boosted variance $(\sigma_i^2)^{\text{new}}$:

- (a) Define $\tilde{\sigma}_i^2 = \sigma_i^2 / C_{i0}$ to move from model (H.3) to model (H.4).
- (b) Resample $(C_{i0})^{\text{new}}$ from the posterior $p(C_{i0} | \tilde{\sigma}_i^2, \cdot)$ obtained under model (H.4):

$$p(C_{i0} | \tilde{\sigma}_i^2, \cdot) \propto p(C_{i0} | b_i^\sigma) \prod_{j: \delta_{ij}=1} p(\beta_{ij} | C_{i0}, \cdot) \prod_{t=1}^T p(\epsilon_{it} | C_{i0}),$$

where $\epsilon_{it} = y_{it} - \beta_{r,i} \mathbf{f}_t^r$. Under any hierarchical shrinkage prior in the slab, this posterior reads:

$$p(C_{i0} | \tilde{\sigma}_i^2, \cdot) \propto (C_{i0})^{a^\sigma - 1} \exp \left\{ -\frac{a^\sigma C_{i0}}{b_i^\sigma} \right\} \left(\frac{1}{C_{i0}} \right)^{\frac{q_i}{2}} \times$$

¹¹The generalized inverse Gaussian distribution, $Y \sim \mathcal{GIG}(p, a, b)$, is a three-parameter family of probability distribution with support $y \in \mathbb{R}^+$. The density is given by

$$f(y) = \frac{(a/b)^{p/2}}{2K_p(\sqrt{ab})} y^{p-1} e^{-(a/2)y} e^{-b/(2y)},$$

where $K_p(z)$ is the modified Bessel function of the second kind, $a > 0$, $b > 0$ and p is a real parameter.

$$\exp \left\{ -\frac{1}{2\tilde{\sigma}_i^2 \kappa C_{i0}} \sum_{j=1}^r \delta_{ij} \frac{\beta_{ij}^2}{\theta_j \omega_{ij}} \right\} \left(\frac{1}{C_{i0}} \right)^{\frac{T}{2}} \exp \left\{ -\frac{1}{2\tilde{\sigma}_i^2 C_{i0}} \sum_{t=1}^T \epsilon_{it}^2 \right\},$$

where $q_i = \sum_{j=1}^r \delta_{ij}$. This yields:

$$(C_{i0})^{\text{new}} | \tilde{\sigma}_i^2, \cdot \sim \mathcal{GIG} \left(a^\sigma - \frac{q_i}{2} - \frac{T}{2}, \frac{2a^\sigma}{b_i^\sigma}, \frac{S_i}{\tilde{\sigma}_i^2} \right),$$

$$S_i = \sum_{t=1}^T \epsilon_{it}^2 + \frac{1}{\kappa} \sum_{j=1}^r \delta_{ij} \frac{\beta_{ij}^2}{\theta_j \omega_{ij}}.$$

For a fractional prior, $p(\beta_{ij} | \delta_{ij} = 1)$ is independent of C_{i0} and the posterior simplifies to:

$$(C_{i0})^{\text{new}} | \tilde{\sigma}_i^2, \cdot \sim \mathcal{GIG} \left(a^\sigma - \frac{T}{2}, \frac{2a^\sigma}{b_i^\sigma}, \frac{1}{\tilde{\sigma}_i^2} \sum_{t=1}^T \epsilon_{it}^2 \right).$$

- (c) Define $(\sigma_i^2)^{\text{new}} = \tilde{\sigma}_i^2 (C_{i0})^{\text{new}} = \sigma_i^2 (C_{i0})^{\text{new}} / C_{i0}$ and move back from model (H.4) to model (H.3).

Appendix I: More about post-processing the posterior draws

I.1 Ensuring variance identification

As a first step in post-processing, we check for each draw δ_r from the CFA model whether the 3579 counting rule is satisfied and the variance decomposition is unique. The efficient algorithm of [Hosszejni and Frühwirth-Schnatter \(2022\)](#) for the verification of the 3579 counting rule is used for this purpose. It is implemented in R and MATLAB and the computer code is publicly available at <https://github.com/hdarjus/sparvaride> and, respectively, <https://github.com/hdarjus/sparvaride-matlab>.

In our experience, the fraction p_V of variance identified draws is relatively high for reasonably chosen priors. This is documented in full detail in Appendix K.3 for the various simulation scenarios considered in Section 5.1. A low fraction of variance identified draws can be the result of poorly chosen shrinkage priors and is a hint to consider another prior family. Using a fractional prior as slab distribution for the loadings β_{ij} , for instance, tends to deliver a low fraction of variance identified draws in high-dimensional factor models, see Section 5.3 for illustration.

I.2 Checking convergence

Functionals of Ω and Σ_r , such as the trace, the log determinant, the sum $\mathbf{1}_m' \Omega^{-1} \mathbf{1}_m$ and the model size d of all variance identified draws are useful means to assess MCMC

convergence. Posterior draws of d are particularly useful to check convergence and assessing efficiency of the MCMC sampler, as d captures the ability of the sampler to move across (variance identified) UGLT factor models of different dimensions.

I.3 Resolving rotational invariance

Due to the UGLT structure imposed on β_r , rotational invariance reduces for variance identified draws to sign and column switching. To rotate β_r into Λ , the columns of β_r are ordered such that the pivots $\mathbf{l}_r = (l_1, \dots, l_r)$ are increasing. Furthermore, the sign is reversed in all columns, where the leading factor loading is negative. The same column permutation is used to reorder any other column-specific posterior draw, in particular the sparsity matrix δ_r , the draws of $\tau_r = (\tau_1, \dots, \tau_r)$ and the draws of the factors $\mathbf{f}_t^r$ for all $t = 1, \dots, T$. Furthermore, in a hierarchical Gaussian shrinkage prior, the column-specific shrinkage parameters $\theta_1, \dots, \theta_r$ have to be reordered as well as the local shrinkage parameters $\omega_{i1}, \dots, \omega_{ir}$ (if any) in every row $i = 1, \dots, m$. Sign switching, however, is not an issue for any of these posterior draws.

I.4 Further inference derived from the GLT draws

Resolving rotational invariance yields posterior draws of loading matrices Λ of varying dimension r that satisfy the GLT condition. The GLT draws Λ very likely exhibit posterior variation in the factor dimension r and in the pivots $l_1 < \dots < l_r$. Among these draws, the highest probability model (HPM), i.e. the sparsity matrix δ_P^Λ visited most often can be determined, and its frequency f_P (an estimator of the posterior probability of the HPM), its factor dimension r_P , its model size d_P , and its sequence of pivots $\hat{\mathbf{l}}_P$ are of interest.

To estimate the factor loading matrix Λ , further inference can be performed conditionally on any estimator $\hat{r}$ of the factor dimension r and a corresponding estimator of the pivots $\hat{\mathbf{l}}_{\hat{r}}$. Bayesian model averaging over all posterior draws of Λ with pivot $\hat{\mathbf{l}}_{\hat{r}}$ yields the posterior mean $E(\Lambda|\mathbf{y}, \hat{\mathbf{l}}_{\hat{r}})$, while uncertainty quantification can be based on the marginal posterior $p(\Lambda_{ij}|\mathbf{y}, \hat{\mathbf{l}}_{\hat{r}})$ of individual elements Λ_{ij} as well as the marginal inclusion probability $\Pr(\delta_{ij}^\Lambda = 1|\mathbf{y}, \hat{\mathbf{l}}_{\hat{r}})$ of individual elements δ_{ij}^Λ in the sparsity matrix δ^Λ of Λ . The corresponding median probability model (MPM) δ_M^Λ is obtained by setting each indicator to one whenever $\Pr(\delta_{ij}^\Lambda = 1|\mathbf{y}, \hat{\mathbf{l}}_{\hat{r}}) \geq 0.5$.

In Section 4.4, we choose the posterior mode $\tilde{r}$ as estimator $\hat{r}$ and the pivots $\hat{\mathbf{l}}_{\tilde{r}} = (\hat{l}_1, \dots, \hat{l}_{\tilde{r}})$ visited most often across all draws with fixed dimension $r = \tilde{r}$, however other estimators can be used, e.g. the factor dimension r_P and the pivot rows $\hat{\mathbf{l}}_P$ of the highest probability model (HPM). Depending on the information in the data, r_P and $\hat{\mathbf{l}}_P$ might be equal to or different from $\tilde{r}$ and $\hat{\mathbf{l}}_{\tilde{r}}$. However, if the posterior frequency f_P of the HPM is small, this might lead to an unstable estimator for Λ .

Across all variance identified draws, the pivots $l_1 < \dots < l_r$ are draws from the marginal posterior distribution $p(l_1, \dots, l_r|\mathbf{y})$, averaged over the varying factor dimension r , whereas all pivots with a fixed number of factors $r = \hat{r}$ are draws from the

conditional posterior distribution $p(l_1, \dots, l_r | r = \hat{r}, \mathbf{y})$. Hence, another estimator for the pivots is the sequence $\mathbf{l}^* = (l_1^*, \dots, l_{r^*}^*)$ visited most often across all variance identified draws with varying dimension r . The corresponding frequency f^* reflects overall posterior uncertainty with respect to choosing the pivots. The number r^* of elements in $\mathbf{l}^*$ provides yet another estimator of the number of factors. If the frequency f^* is small, then the estimator r^* might not coincide with the posterior mode $\tilde{r}$ and, even, if it does, $\mathbf{l}^*$ could be different from $\hat{\mathbf{l}}_{\tilde{r}}$.

Appendix J: MCMC for sparse BFA under a PLT condition

A minor modification of Algorithm 1 allows us to perform sparse BFA under a PLT condition for a *fixed* number r of factors. Due to the PLT condition the r pivots occupy the first r rows, i.e. $(l_1, \dots, l_r) = (1, \dots, r)$. As under the UGLT condition, the loadings below the pivot rows are subject to variable selection. Priors are the same as under UGLT, however the maximum number of factors is equal to $H = r$.

Algorithm 1 is started with the selected number of factors r , no spurious columns and $\mathbf{l}_r = (1, \dots, r)$. Both r and $\mathbf{l}_r$ are held fixed throughout all MCMC iterations and spurious columns that might pop up during variable selection in Step (D) are not removed. This modified sampler skips Step (D-b) and Step (L) in the (CFA) part as well as the entire (EFA) part and consists of the following steps: (H), (D-a), (P), (F), (S) and (A).

All posterior draws exhibit $k = r$ non-zero columns some of which might be spurious. Due to the PLT condition, rotational invariance is not an issue and no reordering of the columns is performed. However, posterior draws that are not variance identified are removed.

Appendix K: More about the simulation study

This section provides more details about the simulation study in Section 5.1.

K.1 Simulation setting

The simulation study assumes $m = 30$, $T = 100$, and $r_{\text{true}} = 5$ factors. We consider six different scenarios regarding the sparsity matrix δ_5 , namely

- (I) a dedicated factor model, where the first 6 variables load on factor 1, the next 6 variables load on factor 2, and so forth, and the final 6 variables load on factor 5;
- (II) a dedicated factor model with overlap, where variables 1 to 7 load on factor 1, variables 6 to 13 load on factor 2, variables 12 to 19 load on factor 3, variables 18 to 25 load on factor 5, and, finally, variables 24 to 30 load on factor 5;

- (III) a two-block factor model, where the first 15 observations load only on factors 1 and 2, while the remaining 15 observations only load on factors 3, 4, and 5;
- (IV) a sparse factor model with 50% sparsity;
- (V) a factor model with a market factor that loads on all measurements, while the remaining four columns exhibit 60% sparsity;
- (VI) and a dense factor model without any zero loadings.

For all models, a GLT condition is imposed on $\mathbf{\Lambda}$, to asses the ability of the various shrinkage priors to recover the six specific sparsity patterns. For four sparsity patterns, the true sparsity pattern δ_5 is deterministic and satisfies the counting rule for variance identification by design. The pivots are given by $\mathbf{l}_5 = (1, 7, 13, 19, 25)$ for a dedicated factor model, $\mathbf{l}_5 = (1, 6, 12, 18, 24)$ for a dedicated factor model with overlap, $\mathbf{l}_5 = (1, 2, 16, 17, 18)$ for a block-factor model and $\mathbf{l}_5 = (1, 2, 3, 4, 5)$ for a dense factor model.

For the remaining two settings, the true sparsity matrix δ_5 is random. The first pivot $l_1 = 1$, while the remaining pivots $1 < l_2 < l_3 < l_4 < l_5$ obey a GLT structure and are drawn randomly without replacement from row 2 to 20 and ordered by size. With the level of sparsity denoted by $p_s = \Pr(\delta_{ij} = 0)$, all loadings below the pivot rows are set to zero with probability p_s , i.e. $\delta_{ij} = \mathbb{I}(U_{ij} \leq 1 - p_s)$, where U_{ij} is a uniform random number. Note that $p_s = 0.5$ for a sparse factor model and $p_s = 0.6$ for a market-factor model. In both settings, the simulated sparsity matrix δ_5 might violate the counting rule for variance identification. To ensure variance identification for each data set, simulations are repeated until a sparsity matrix δ_5 is obtained that satisfies the counting rule for variance identification.

For all six scenarios, non-zero factor loadings are drawn as $\lambda_{ij} = (-1)^{b_{ij}} \mathcal{N}(1, 0.1)$, where the exponent b_{ij} is a binary variable with $\Pr(b_{ij} = 1) = 0.1$. Finally, for all scenarios the idiosyncratic variances are assumed to be homogeneous with $\sigma_i^2 \equiv 1$ for $i = 1, \dots, m$.

Depending on the scenario, the row size $q_i = \sum_{j=1}^5 \delta_{ij}$ takes values between 1 to 5. The average expected row size over the 30 rows, denoted by $E(q_i)$, varies for the six scenarios and ranges from $E(q_i) = 1$ for a dedicated factor model, $E(q_i) = 1.26$ for a dedicated factor model with overlap, and $E(q_i) = 1.92$ for a sparse factor model to $E(q_i) = 2.15$ for a market-factor model, $E(q_i) = 2.37$ for a two-block factor model, and $E(q_i) = 4.67$ for a dense factor model.

The average of the expected communalities

$$E \left(\frac{\sum_{j=1}^5 \beta_{ij}^2}{\sum_{j=1}^5 \beta_{ij}^2 + \sigma_i^2} \right)$$

over the 30 rows, denoted by $E(R_i^2)$, varies across the six scenarios and ranges from $E(R_i^2) = 0.48$ for a dedicated factor model and $E(R_i^2) = 0.53$ for a dedicated factor model with overlap to $E(R_i^2) = 0.57$ for a sparse factor model, $E(R_i^2) = 0.64$ for a market-factor model, $E(R_i^2) = 0.69$ for a two-block factor model, and $E(R_i^2) = 0.82$ for a dense factor model.

K.2 Performance measures

The variance identified posterior draws with varying dimension r are used to estimate various performance measures for each data set.

Regarding the covariance matrices, we consider the posterior expectation $E(L(\mathbf{\Omega}_r, \mathbf{\Omega}_0)|\mathbf{y})$ of the entropy (or Stein) loss (Yang and Berger, 1994) in recovering the true covariance matrix $\mathbf{\Omega}_0$ of the data:

$$L(\mathbf{\Omega}_r, \mathbf{\Omega}_0) = \text{tr}(\mathbf{\Omega}_r \mathbf{\Omega}_0^{-1}) - \log |\mathbf{\Omega}_r \mathbf{\Omega}_0^{-1}| - m,$$

the posterior expectation $E(L(\mathbf{\Omega}_r^{-1}, \mathbf{\Omega}_0^{-1})|\mathbf{y})$ of the entropy loss for $\mathbf{\Omega}_0^{-1}$:

$$L(\mathbf{\Omega}_r^{-1}, \mathbf{\Omega}_0^{-1}) = \text{tr}(\mathbf{\Omega}_r^{-1} \mathbf{\Omega}_0) - \log |\mathbf{\Omega}_r^{-1} \mathbf{\Omega}_0| - m,$$

as well as the posterior expectation $E(L(\mathbf{\Sigma}_r, \mathbf{\Sigma}_0)|\mathbf{y})$ of the entropy loss in recovering the true covariance matrix $\mathbf{\Sigma}_0 = \text{Diag}(\sigma_{0,1}^2, \dots, \sigma_{0,m}^2)$ of the idiosyncratic errors:

$$L(\mathbf{\Sigma}_r, \mathbf{\Sigma}_0) = \sum_{i=1}^m \left(\frac{\sigma_i^2}{\sigma_{0,i}^2} - \log \frac{\sigma_i^2}{\sigma_{0,i}^2} - 1 \right).$$

To assess recovery of the true sparsity pattern $\boldsymbol{\delta}^\Lambda = \boldsymbol{\delta}_5$, we consider the bias in d_r compared to the true model size d_{true} ,

$$B_d = E(d_r|\mathbf{y}) - d_{\text{true}}, \quad d_r = \sum_{i,j} \delta_{ij}, \quad d_{\text{true}} = \sum_{i,j} \delta_{ij}^\Lambda,$$

as well as the true positive rate TP_Ω for non-zero and the false discovery (false positive) rate FP_Ω for zero correlations in $\mathbf{\Omega}_0$,

$$\begin{aligned} \text{TP}_\Omega &= E \left(\frac{1}{c_{\text{true}}} \sum_{i,\ell < i} \mathbb{I}(\mathbf{\Omega}_{r,i\ell} \neq 0) \mathbb{I}(\mathbf{\Omega}_{0,i\ell} \neq 0) \middle| \mathbf{y} \right), \quad c_{\text{true}} = \sum_{i,\ell < i} \mathbb{I}(\mathbf{\Omega}_{0,i\ell} \neq 0), \\ \text{FP}_\Omega &= E \left(\frac{1}{c_r} \sum_{i,\ell < i} \mathbb{I}(\mathbf{\Omega}_{r,i\ell} \neq 0) \mathbb{I}(\mathbf{\Omega}_{0,i\ell} = 0) \middle| \mathbf{y} \right), \quad c_r = \sum_{i,\ell < i} \mathbb{I}(\mathbf{\Omega}_{r,i\ell} \neq 0). \end{aligned}$$

Since for any of these measures the factor dimension r in the underlying posterior draws is varying, these measures incorporate the uncertainty in recovering the true factor dimension r .

For all data sets, where the posterior mode $\tilde{r} = r_{\text{true}}$ is equal to the true number of factors, the sparsity pattern $\boldsymbol{\delta}_r$ of all posterior draws with $r = r_{\text{true}}$ is compared with the true sparsity pattern $\boldsymbol{\delta}^\Lambda$ through the true positive rate TP_δ and the false discovery (false positive) rate FP_δ , defined by:

$$\text{TP}_\delta = E \left(\frac{1}{d_{\text{true}}} \sum_{i,j} \delta_{ij} \delta_{ij}^\Lambda \middle| r = r_{\text{true}}, \mathbf{y} \right),$$

$$\text{FP}_\delta = \text{E} \left(\frac{1}{d_r} \sum_{i,j} \delta_{ij} (1 - \delta_{ij}^\Lambda) \middle| r = r_{\text{true}}, \mathbf{y} \right).$$

Note that the pivots in δ_r very likely are still varying, even if r is equal to the correct number of factors. Hence, these measures also incorporate the uncertainty in recovering the true pivot rows of δ^Λ .

K.3 Additional results

Table K.1 complements Table 2 and reports averages for two additional slab distributions, namely prior (3.11) with column-specific shrinkage (C), and prior (3.12), where local shrinkage with $a^\omega = c^\omega = 0.5$ relies on the horseshoe (H).

The top eight panels of Figures K.2 to K.7 provide for each sparsity setting box plots of the sampling distribution of the various performance measures for sparse Bayesian factor analysis based on the UGLT condition under all five slab distributions P_{slab} for β_{ij} , namely a fractional prior (F), prior (3.10) with global shrinkage (G), prior (3.11) with column-specific shrinkage (C), and prior (3.12), where local shrinkage with $a^\omega = c^\omega = 0.5$ relies on the horseshoe (H) and on a triple gamma with $a^\omega = c^\omega = 0.2$ (T).

The bottom eight panels of Figures K.2 to K.7 compare these performance measures to the corresponding measures obtained for sparse Bayesian factor analysis under the PLT condition, if $r = 5$ is assumed to be known and equal to the true number of factors.

Table K.1: Performance of sparse UGLT Bayesian factor analysis for six sparsity patterns under homogeneous errors ($r_{\text{true}} = 5$) for various shrinkage priors; prior (3.14) for σ_i^2 .

		$\tilde{r}$	P_{true}	R_{Ω}	$R_{\Omega-1}$	R_{Σ}	TP_{Ω}	FP_{Ω}	B_d	TP_{δ}	FP_{δ}
Dedicated											
UGLT	C	5.02(0,1)	0.935	1.48	1.59	0.92	97.9	39.8	8.23	96.4	22.9
	H	5.02(0,1)	0.935	1.54	1.63	1	97.2	52.7	13.6	95.9	32.5
PLT	C	-	-	2.3	2.14	1.14	93.5	58.6	14.7	38.7	73.9
	H	-	-	2.57	2.4	1.46	91.9	69.6	23.2	38.2	78.6
Overlap											
UGLT	C	5	0.931	1.68	1.77	0.89	97.3	34.4	9.66	97.5	21.1
	H	5.02(0,1)	0.942	1.74	1.83	0.96	97.7	46.8	16.5	97.4	30.9
PLT	C	-	-	2.35	2.33	1.06	95.4	50.6	16.6	49.4	65.6
	H	-	-	2.53	2.49	1.15	96.3	62.1	29.3	55	68.9
Block											
UGLT	C	4.86(7,1)	0.805	2.51	2.52	1.01	95.0	25.0	-7.89	79.2	12.9
	H	4.88(6,1)	0.836	2.54	2.56	1.09	97.0	36.7	0.21	81.3	19.9
PLT	C	-	-	3.32	3.69	1.23	90.9	37.5	-6.61	54.2	40.7
	H	-	-	3.60	3.50	1.45	95.0	47.2	8.86	62.3	44.8
Sparse											
UGLT	C	4.94(1,2)	0.912	2.37	2.48	0.9	96.8	25.9	17.1	95.3	25.6
	H	4.96(2,0)	0.911	2.44	2.52	1	99.4	37.7	30	95.2	37.3
PLT	C	-	-	2.65	2.71	0.92	98.1	34.1	21.3	73.4	46.5
	H	-	-	2.91	2.99	1.07	99	37.1	36.9	78.7	51.5
Market											
UGLT	C	4.88 (4,1)	0.889	2.28	2.39	0.93	99.6	0	10.5	96.1	17.1
	H	4.94 (3,0)	0.921	2.55	2.62	1.02	99.7	0	20.5	95.9	26.9
PLT	C	-	-	2.54	2.75	0.92	99.3	0	15.8	77.9	37.9
	H	-	-	2.76	3.43	1.06	97.8	0	25.6	82.4	42.1
Dense											
UGLT	C	5	0.972	3.76	3.63	0.9	99.2	0	-23.7	83.0	0
	H	4.96(2,0)	0.942	4.13	3.94	0.99	99.5	0	-16.7	88.7	0
PLT	C	-	-	3.85	3.73	0.92	98.9	0	-26.1	81.3	0
	H	-	-	4.21	3.93	1	99.3	0	-16.4	88.3	0

For each performance measure, the average across the 50 simulated data sets is reported. If $\tilde{r} \neq 5$, (a,b) report, respectively, cases of under- and overfitting.

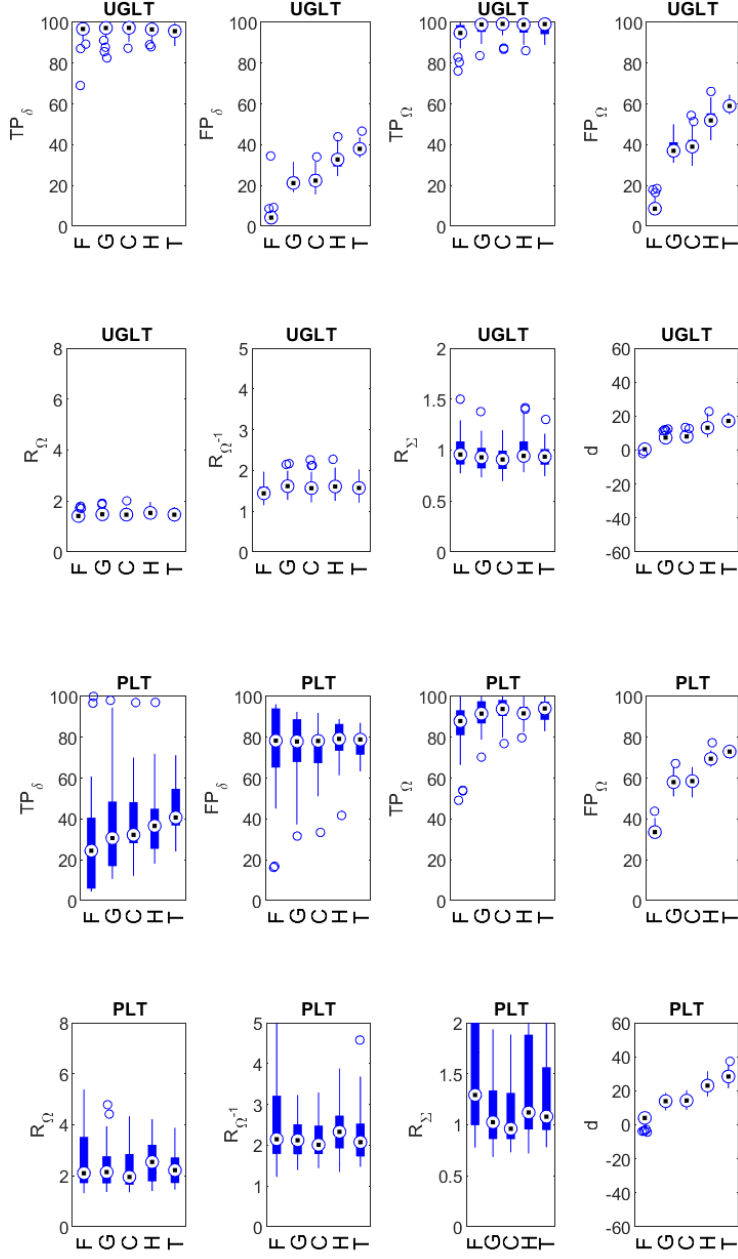


Figure K.2: Performance of sparse Bayesian factor analysis under a UGLT condition with r unknown (top) in comparison to a PLT condition with $r = r_{\text{true}} = 5$ known (bottom) for the following sparsity pattern: dedicated factor model.

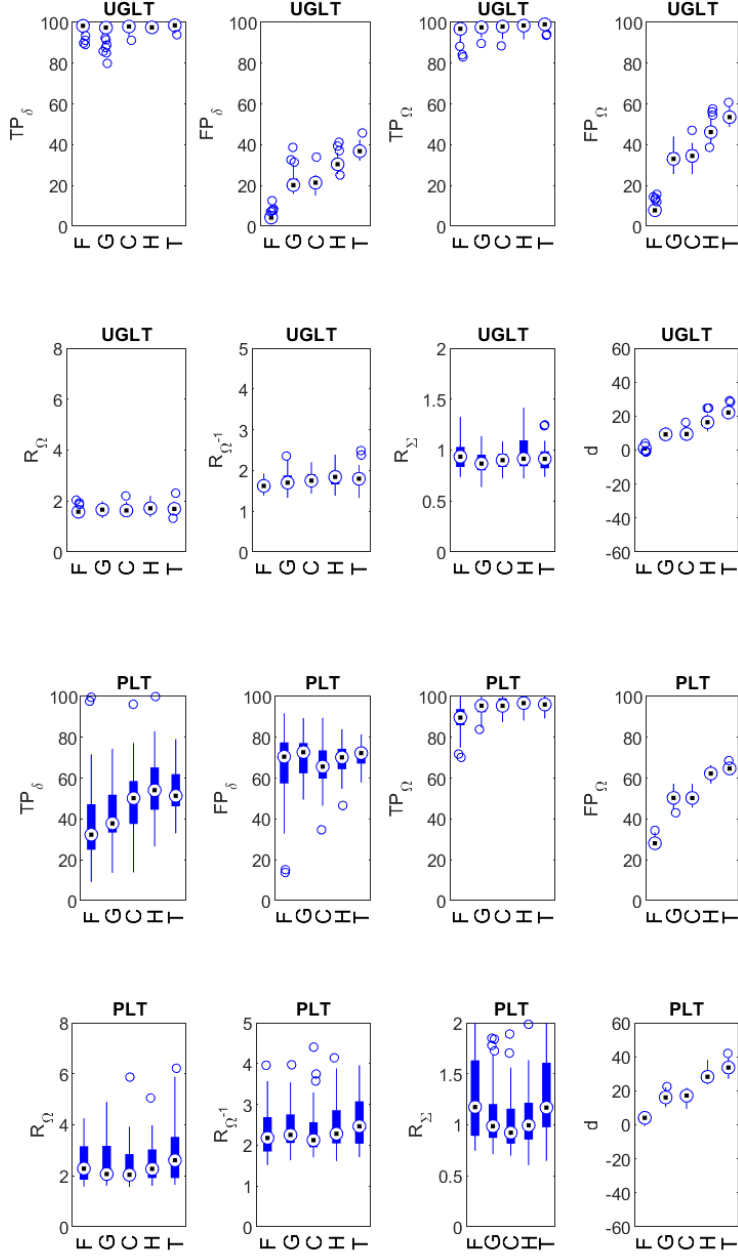


Figure K.3: Performance of sparse Bayesian factor analysis under a UGLT condition with r unknown (top) in comparison to a PLT condition with $r = r_{\text{true}} = 5$ known (bottom) for the following sparsity pattern: dedicated factor model with overlap.

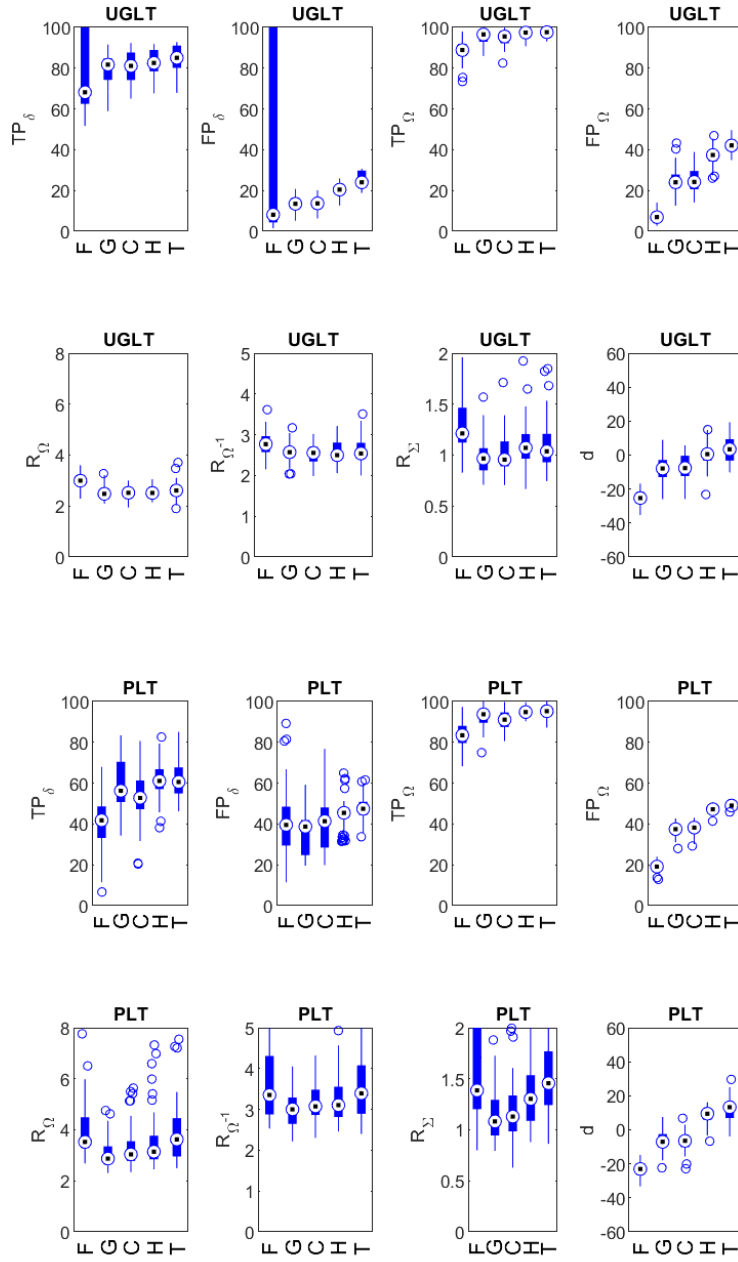


Figure K.4: Performance of sparse Bayesian factor analysis under a UGLT condition with r unknown (top) in comparison to a PLT condition with $r = r_{\text{true}} = 5$ known (bottom) for the following sparsity pattern: block-factor model.

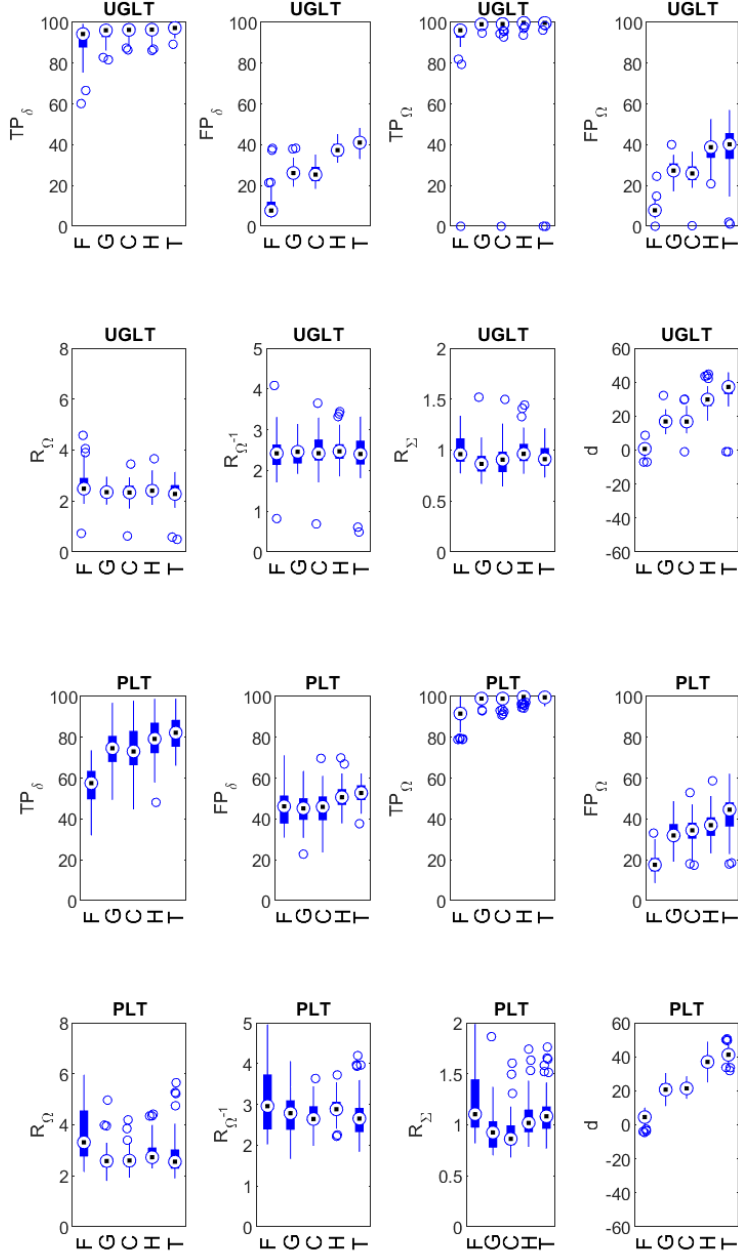


Figure K.5: Performance of sparse Bayesian factor analysis under a UGLT condition with r unknown (top) in comparison to a PLT condition with $r = r_{\text{true}} = 5$ known (bottom) for the following sparsity pattern: sparse factor model.

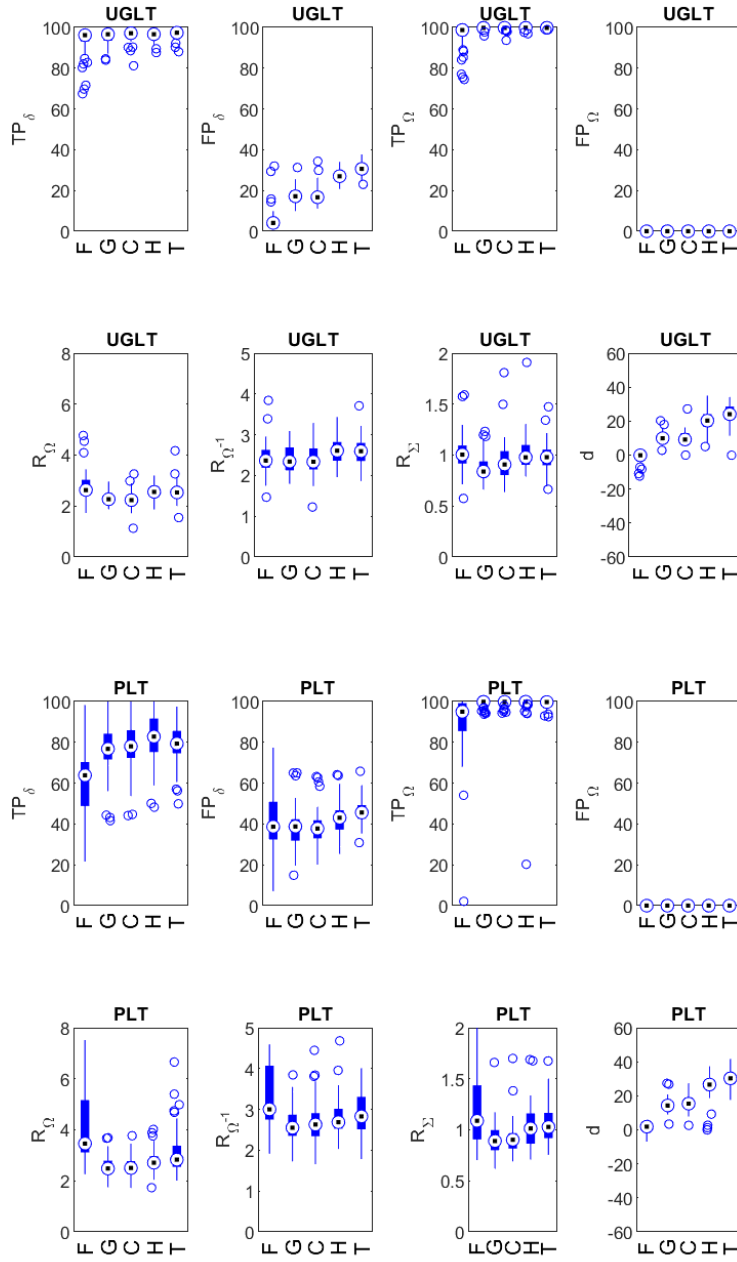


Figure K.6: Performance of sparse Bayesian factor analysis under a UGLT condition with r unknown (top) in comparison to a PLT condition with $r = r_{\text{true}} = 5$ known (bottom) for the following sparsity pattern: market-factor model.

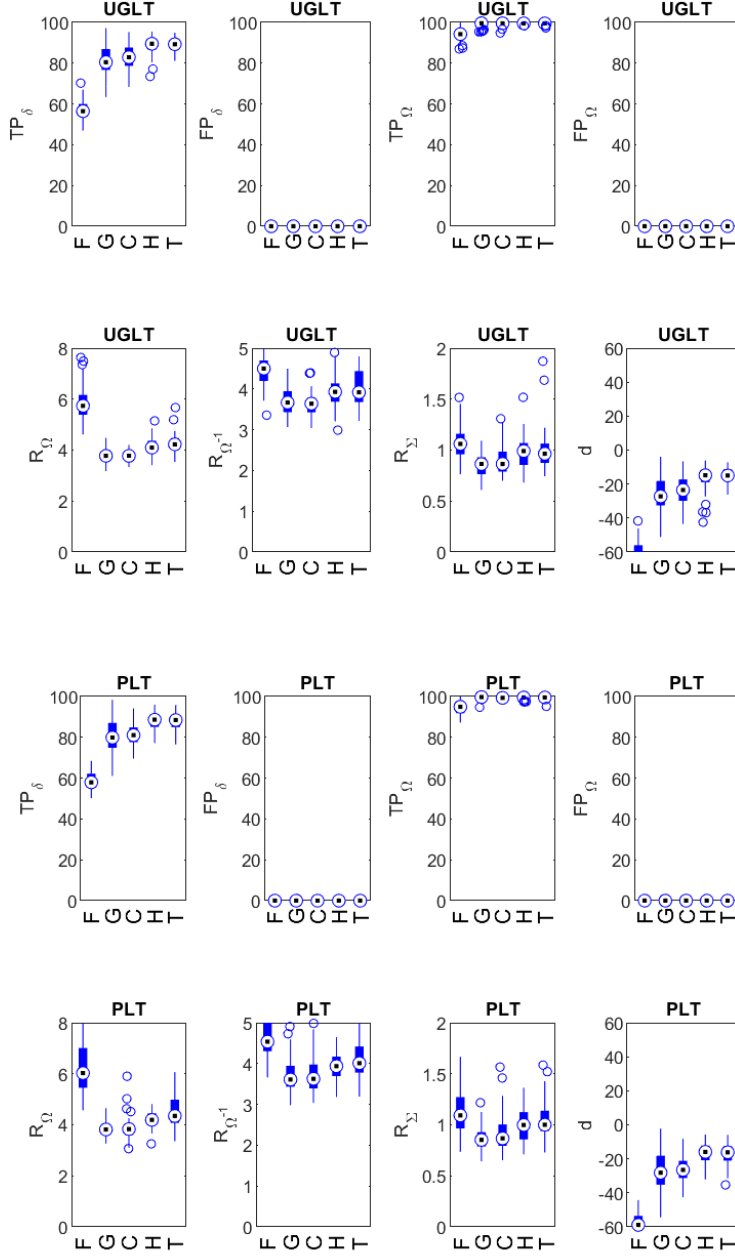


Figure K.7: Performance of sparse Bayesian factor analysis under a UGLT condition with r unknown (top) in comparison to a PLT condition with $r = r_{\text{true}} = 5$ known (bottom) for the following sparsity pattern: dense factor model.

Appendix L: More details for the exchange rate data

L.1 Data

The 22 exchange rates analysed in Section 5.2 are listed in Table L.2.

1	AUD	Australian dollar	12	NZD	New Zealand dollar
2	CAD	Canadian dollar	13	PHP	Philippine peso
3	CZK	Czech koruna	14	PLN	Polish złoty
4	DKK	Danish krone	15	RON	Romanian fourth leu
5	HKD	Hong Kong dollar	16	RUB	Russian ruble
6	IDR	Indonesian rupiah	17	SEK	Swedish krona
7	JPY	Japanese yen	18	CHF	Swiss franc
8	KRW	South Korean won	19	SGD	Singapore dollar
9	MXN	Mexican peso	20	THB	Thai baht
10	MYR	Malaysian ringgit	21	GBP	(U.K.) pound sterling
11	NOK	Norwegian krone	22	USD	U.S. dollar

Table L.2: Exchange rate data; currency abbreviations.

L.2 Further results

Additional posterior summaries for the ordered GLT draws (with r still being a varying factor dimension) are reported in Table L.3 for all four priors. $\mathbf{1}^* = (1, 2, 5, 7)$ is the most likely sequence of pivots across varying dimension, confirming the presence of four factors, and is equal to $\hat{\mathbf{l}}_4$ derived from all posterior draws with $r = \tilde{r} = 4$ factors.

Additional results are provided for the HIG prior on σ_i^2 . Table L.4 reports $\Pr(\delta_{ij} = 1 | \mathbf{y}, \hat{\mathbf{l}}_4)$, averaged over all ordered GLT draws with pivots equal to $\hat{\mathbf{l}}_4 = (1, 2, 5, 7)$. These probabilities are used to derive the median probability model (MPM) which coincides with the sparsity matrix of the highest probability model (HPM) for all priors, see the left-hand side of Figure 5 for illustration.

Table L.5 shows the posterior mean of the factor loading matrix, the idiosyncratic variances and the communalities, obtained by averaging over all draws where the pivots of δ_4 coincide with $\hat{\mathbf{l}}_4 = (1, 2, 5, 7)$ for the HIG prior on σ_i^2 . As expected, non-zero factor loadings have relatively high communalities for the different currencies, whereas for zero rows the communalities are practically equal to zero percent. The posterior mean of the factor loading matrix is also depicted in the right-hand side of Figure 5.

Prior on σ_i^2	N_v	$100f_P$	$\mathbf{l}_P$	r_P	d_P	$\mathbf{l}^*$	$100f^*$	r^*	d_M
HF	19332	4.1	(1,2,5,7)	4	26	(1,2,5,7)	78.0	4	26
HIG	21747	3.2	(1,2,5,7)	4	26	(1,2,5,7)	80.6	4	26
FSL	18692	4.8	(1,2,5,7)	4	26	(1,2,5,7)	88.5	4	26
BD	20005	4.2	(1,2,5,7)	4	26	(1,2,5,7)	79.8	4	26

Table L.3: Exchange rate data (standardized); total number of visited models N_v ; frequency $100f_P$ (in percent), pivots $\mathbf{l}_P$, number of factors r_P and model size d_P of the HPM; pivots $\mathbf{l}^*$ visited most often, corresponding frequency f^* (in percent) and number of factors r^* ; model size d_M of the MPM.

Currency	Factor 1	Factor 2	Factor 3	Factor 4
AUD	1	0	0	0
CAD	1	1	0	0
CZK	0.01	0.11	0	0
DKK	0	0.19	0	0
HKD	0.01	1	1	0
IDR	0.05	1	0.05	0
JPY	0.01	1	0.96	1
KRW	0.02	1	0.08	1
MXN	0.09	1	0.01	0.02
MYR	0	1	0.08	0.02
NOK	0	0.15	0.01	0.02
NZD	1	0.06	0.01	0.01
PHP	0	1	0.01	0.03
PLN	0.06	0.43	0.06	0.02
RON	0	1	0.93	0.10
RUB	0	1	0.01	0.75
SEK	0.09	0.06	0.34	0.01
CHF	0.23	0.1	0.08	0.09
SGD	0.01	1	0.01	1
THB	0.02	1	0.04	0.99
GBP	0	1	0.02	0.02
USD	0.01	1	1	0.01

Table L.4: Exchange rate data (standardized); inclusion probabilities for the sparsity matrix δ_4 , averaged over all ordered GLT draws with pivots equal to $\hat{\mathbf{l}}_4 = (1, 2, 5, 7)$ (fractional prior combined with the HIG prior on σ_i^2).

Currency	Factor loadings				Communalities				
	Λ_{i1}	Λ_{i2}	Λ_{i3}	Λ_{i4}	R_{i1}^2	R_{i2}^2	R_{i3}^2	R_{i4}^2	σ_i^2
AUD	0.96	0	0	0	88.1	0	0	0	0.122
CAD	0.39	0.605	0	0	16.9	39.0	0	0	0.416
CZK	≈ 0	-0.014	0	0	≈ 0	0.32	0	0	0.968
DKK	≈ 0	0.0359	0	0	≈ 0	0.90	0	0	0.964
HKD	≈ 0	1.07	0.196	0	≈ 0	94.8	3.37	0	0.021
IDR	0.01	0.572	-0.010	0	0.28	31.8	0.27	0	0.7
JPY	≈ 0	0.504	0.415	0.770	≈ 0	21.9	15.6	50.4	0.141
KRW	≈ 0	0.803	-0.016	0.427	≈ 0	57.6	0.36	16.9	0.278
MXN	0.015	0.929	≈ 0	≈ 0	0.26	75.7	≈ 0	≈ 0	0.269
MYR	≈ 0	1.07	≈ 0	≈ 0	≈ 0	95.1	≈ 0	≈ 0	0.057
NOK	≈ 0	0.025	≈ 0	≈ 0	≈ 0	0.6	≈ 0	≈ 0	0.967
NZD	0.795	≈ 0	≈ 0	≈ 0	61.6	≈ 0	≈ 0	≈ 0	0.389
PHP	≈ 0	0.889	≈ 0	≈ 0	≈ 0	69.9	≈ 0	≈ 0	0.336
PLN	0.017	0.111	-0.014	≈ 0	0.52	3.31	0.44	≈ 0	0.939
RON	≈ 0	0.555	-0.492	0.021	≈ 0	29.1	23.8	0.57	0.493
RUB	≈ 0	1	≈ 0	0.126	≈ 0	85.5	≈ 0	1.98	0.144
SEK	0.026	≈ 0	-0.123	≈ 0	0.84	0.1	4.9	≈ 0	0.92
CHF	-0.08	0.014	0.025	≈ 0	2.95	0.35	0.85	1.96	0.93
SGD	≈ 0	0.976	≈ 0	0.296	≈ 0	81.5	≈ 0	7.86	0.122
THB	≈ 0	0.751	≈ 0	0.385	≈ 0	51.2	0.15	14.2	0.375
GBP	≈ 0	0.589	≈ 0	≈ 0	≈ 0	33.5	≈ 0	≈ 0	0.69
USD	≈ 0	1.07	0.195	≈ 0	≈ 0	94.9	3.34	≈ 0	0.021

Note: entries with $|\Lambda_{ij}| < 0.01$ and entries with $R_{ij}^2 < 0.1$ are indicated by ≈ 0 .

Table L.5: Exchange rate data (standardized); posterior mean of the factor loadings Λ_{ij} , the communalities R_{ij}^2 (in percent) and the idiosyncratic variances σ_i^2 for a 4-factor model with the GLT constraint $\hat{\mathbf{I}}_4 = (1, 3, 5, 7)$ (fractional prior combined with the HIG prior on σ_i^2).

Appendix M: Adding a non-zero mean

If data $\mathbf{y}_t^N$ are not standardized, then the basic factor model is extended to include a non-zero mean, i.e. the true underlying factor model takes the form

$$\begin{aligned}\mathbf{f}_t &\sim \mathcal{N}_r(\mathbf{0}, \mathbf{I}_r), \\ \mathbf{y}_t^N &= \boldsymbol{\mu} + \boldsymbol{\Lambda} \mathbf{f}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}_m(\mathbf{0}, \boldsymbol{\Sigma}_0), \quad \boldsymbol{\Sigma}_0 = \text{Diag}(\sigma_1^2, \dots, \sigma_m^2),\end{aligned}$$

and implies that, marginally, $\mathbf{y}_t^N$ arises from a multivariate normal distribution with mean vector $\boldsymbol{\mu}$ and a covariance matrix $\boldsymbol{\Omega}$ with factor structure:

$$\mathbf{y}_t^N \sim \mathcal{N}_m(\boldsymbol{\mu}, \boldsymbol{\Omega}), \quad \boldsymbol{\Omega} = \boldsymbol{\Lambda} \boldsymbol{\Lambda}' + \boldsymbol{\Sigma}_0. \quad (\text{M.1})$$

The methods introduced in this paper can be easily extended to such a setting by adding the unknown mean $\boldsymbol{\mu}$ to the EFA model (1.3):

$$\mathbf{y}_t^N = \boldsymbol{\mu} + \boldsymbol{\beta}_H \mathbf{f}_t^H + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}_m(\mathbf{0}, \boldsymbol{\Sigma}_H), \quad \mathbf{f}_t^H \sim \mathcal{N}_H(\mathbf{0}, \mathbf{I}_H).$$

Conditional on the mean vector $\boldsymbol{\mu}$, the innovations $\mathbf{y}_t = \mathbf{y}_t^N - \boldsymbol{\mu}$ are simply treated as input for a factor model with zero mean and no modification regarding the prior specifications are needed.

The mean vector $\boldsymbol{\mu}$ is sampled during MCMC, exploiting the marginal distribution given in (M.1). The corresponding likelihood is combined with a Gaussian prior $\boldsymbol{\mu} \sim \mathcal{N}_m(\mathbf{m}_0, \mathbf{M}_0)$ and yields the posterior:

$$\begin{aligned}\boldsymbol{\mu} | \mathbf{y}, \boldsymbol{\Omega} &\sim \mathcal{N}_m(\mathbf{M}_T \mathbf{m}_T, \mathbf{M}_T), \\ \mathbf{m}_T &= \mathbf{M}_0^{-1} \mathbf{m}_0 + \boldsymbol{\Omega}^{-1} \sum_{t=1}^T \mathbf{y}_t^N, \\ \mathbf{M}_T^{-1} &= \mathbf{M}_0^{-1} + T \boldsymbol{\Omega}^{-1},\end{aligned} \quad (\text{M.2})$$

where $\mathbf{y}_t^N$ are the observed data and $\boldsymbol{\Omega} = \boldsymbol{\beta}_H \boldsymbol{\beta}_H' + \boldsymbol{\Sigma}_H$ is derived from the current posterior draws. Algorithm 1 is extended by adding a Step (M), where $\boldsymbol{\mu}$ is sampled from the Gaussian posterior given in (M.2). Since we marginalized w.r.t. the factors $\mathbf{f}_t^H$ for all $t = 1, \dots, T$, new factors $\mathbf{f}_t^H$ have to be sampled for all $t = 1, \dots, T$ immediately after sampling $\boldsymbol{\mu}$. Hence, we perform Step (M) after Step (P) and immediately before Step (F).

Alternatively, one could sample $\boldsymbol{\mu} | \mathbf{y}, \mathbf{f}_t^H, \cdot$ conditional on the factors as in [Murphy et al. \(2020\)](#), however, we experienced considerable efficiency loss for such a sampler compared to Step (M).

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